Shape Optimization of an Airfoil for Horizontal Axis Wind Turbines in the Coastal Region of Bangladesh.

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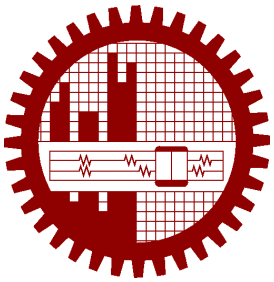
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DECLARATION

This is to certify that the work presented in the thesis is carried out by the authors under the supervision of Dr. Laboni Afroz, Department of Naval Architecture and Marine Engineering, Dhaka. No portion of the work contained in this thesis has been submitted in support of an application for another degree or qualification of this or any other university or institution of learning.

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Dedicated to

Our Beloved Parents and

Respected Teachers.

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ABSTRACT

Low wind speed in the coastal region of Bangladesh limits the amount of energy that can be extracted by conventional wind turbines. This study aimed to improve the aerodynamic performance of the NACA 4412 airfoil for operation under such conditions. The analysis was conducted at a wind speed of 5 m/s and a Reynolds number of approximately 3.42×10^5, covering angles of attack from -5^0 to 15^0. The airfoil geometry was represented by twelve Class Shape Transformation coefficients, which were used as design variables in a Genetic Algorithm. XFOIL was integrated with the optimization process to evaluate the lift-to-drag ratio of each generated airfoil. The final optimized profile was compared with the baseline airfoil using two-dimensional CFD analysis in ANSYS Fluent. Their aerodynamic data were then applied to the same three-bladed rotor model in QBlade to evaluate rotor-level performance. At the design angle of 7^0, the maximum lift-to-drag ratio increased from 95.71 for the baseline airfoil to 119.93 for the optimized airfoil, representing an improvement of 25.3%. The optimized profile had greater camber and lower thickness and produced a higher lift coefficient over the investigated angle range. In QBlade, the optimized rotor achieved a 16.67% increase in power coefficient at a tip speed ratio of 4, together with higher torque and thrust coefficients. The findings suggest that the proposed CST-GA framework can improve airfoil and rotor performance near the selected low-wind-speed design condition.

Keywords: NACA 4412; low wind speed; airfoil optimization; Class Shape Transformation; Genetic Algorithm; CFD; QBlade

Table of Contents

CHAPTER 1: INTRODUCTION 1

1.1 Background Study 1

1.2 Motivation 3

1.3 Objectives 4

1.4 Significance of the Research 5

1.5 Chapter Outline 5

CHAPTER 2: LITERATURE REVIEW 5

2.1 Literature Review 6

2.2 Research Gap 11

CHAPTER 3: METHODOLOGY 12

3.1 Selection of the Baseline Airfoil 13

3.1.1 Airfoil Geometry 13

3.1.2 Airfoil Design 14

3.2 Airfoil Parameterization Using CST 15

3.2.1 Mathematical Model of CST 15

3.2.2 Implementation of CST Parameterization 18

3.2.3 Calculate CST coefficients 19

3.3 Optimization Using a Genetic Algorithm 21

3.3.1 Optimization Scheme 21

3.3.2 Genetic Algorithm 22

3.3.3 GA Parameters 23
3.3.4 Selection of GA parameters 27
3.4 Numerical Analysis 28
3.4.1 Computational model 28
3.4.2 Turbulence modeling 28
3.4.3 Computational domain 30
3.5 Numerical setup and Boundary conditions 31
3.6 Numerical Setup Validation 33
3.7 Rotor Power Performance Analysis 34
3.7.1 Background of Power Calculation 34
3.7.2 Blade Geometry 35
3.7.3 Construction of blades 37
CHAPTER 4: RESULT AND DISCUSSION 39
4.1 GA convergence and selected configuration 39
4.2 Geometry comparison 40
4.3 Aerodynamic performance comparison 41
4.4 CFD verification and Flow pattern comparison 43
4.5 QBlade Power analysis 45
CHAPTER 5: CONCLUSION AND FUTURE SCOPES 48
5.1 Key Findings 48
5.2 Limitations and Future works 49
BIBLIOGRAPHY 50

LIST OF FIGURES

Figure 1.1 Global wind power capacity and Wind energy market size (USD Billion) 1

Figure 1.2 Monthly Average Wind Speed Variation in coastal region of Bangladesh..... 3

Figure 1.3 Coastal zone of Bangladesh..... 3

Figure 1.4 Flow separation on turbine blades..... 3

Figure 1.5 Potential wind firm in the coastal region of Bangladesh..... 4

Figure 3.1 Typical airfoil geometry……13

Figure 3.2 NACA 4412 geometry form UIUC database……14

Figure 3.3 CST parameterization workflow……...18

Figure 3.4 Comparison between baseline and CST generation airfoil……...20

Figure 3.5 Fitting error distribution……20

Figure 3.6 Genetic Algorithm workflow……23

Figure 3.7 Exchange genes among parents to form offspring……24

Figure 3.8 Optimization workflow……26

Figure 3.9 Computational domain blocks……...30

Figure 3.10 Refined meshing near the airfoil edge……30

Figure 3.11 Validation of numerical setup……33

Figure 3.12 Turbine blade geometry……...35

Figure 3.13 Blade geometry for optimized section……37

Figure 3.14 Three blade turbine model……...38

Figure 4.1 Convergence plot..... 39

Figure 4.2 Geometry comparison of baseline and optimized airfoil 40

Figure 4.3 Lift coefficient comparison of baseline and optimized airfoil 41

Figure 4.4 Drag coefficient comparison of baseline and optimized airfoil 41

Figure 4.5 Lift to Drag coefficient ratio comparison of baseline and optimized airfoil..... 42

Figure 4.6 Lift coefficient predicted by XFOIL and ANSYS Fluent for both sections..... 43

Figure 4.7 Velocity-magnitude fields of NACA 4412 (left) and the optimized section (right) at 7° AOA..... 44

Figure 4.8 Static-pressure fields of NACA 4412 (left) and the optimized section (right) at 7° AOA 44

Figure 4.9 Velocity-magnitude fields of NACA 4412 (left) and the optimized section (right) at 13° AOA..... 45

Figure 4.10 Static-pressure fields of NACA 4412 (left) and the optimized section (right) at 13° AOA..... 45

Figure 4.11 QBlade power coefficient versus tip speed ratio for baseline and optimized blade section 46

Figure 4.12 QBlade thrust coefficient versus tip speed ratio..... 47

Figure 4.13 QBlade torque coefficient versus tip speed ratio..... 47

LIST OF TABLES

Table 3. 1 CST coefficients for optimization..... 19

Table 3. 2 Selected GA parameters..... 27

Table 3. 3 Turbulence model parameters..... 30

Table 3. 4 Operating parameters for the airfoil..... 31

Table 3. 5 Solution scheme settings..... 31

Table 3. 6 Normalized chord distribution..... 38

Table 3.7 Design parameters for power calculation……38

Table 4.1 Comparison of geometric parameters……41

NOMENCLATURE AND ABBREVIATIONS

NACA	National Advisory Committee for Aeronautics
C	Class function
C_d	Drag coefficient
C_l	Lift coefficient
C_P	Pressure coefficient
CST	Class shape transformation
D	Drag force [N]
GA	Genetic algorithm
K	Binomial coefficient
L	Lift force [N]
L/D	Lift to drag ratio
N	Order of Bernstein polynomial
P_c	Crossover Probability
P_m	Mutation Probability
P_S	Population size

RANS	Reynolds Averaged Navier-Stokes's equation
Re	Reynolds number
S	Shape function
TE	Trailing edge
LE	Leading edge

CHAPTER 1: INTRODUCTION

1.1 Background Study

The growing demand for energy and the environmental problems related to fossil fuels have increased the importance of renewable energy. Among the available renewable sources, wind energy has become an important option because it is clean, widely available and capable of producing electricity without direct greenhouse-gas emissions during operation. The aerodynamic performance of a wind turbine mainly depends on the blade geometry, and the airfoil section is one of the most important parts of this geometry. An airfoil that can generate higher lift with lower drag allows the blade to extract wind energy more efficiently. Therefore, aerodynamic shape optimization of airfoils is an important step in improving wind-turbine performance [1], [2].

Airfoil optimization requires a suitable method to represent the geometry with a limited number

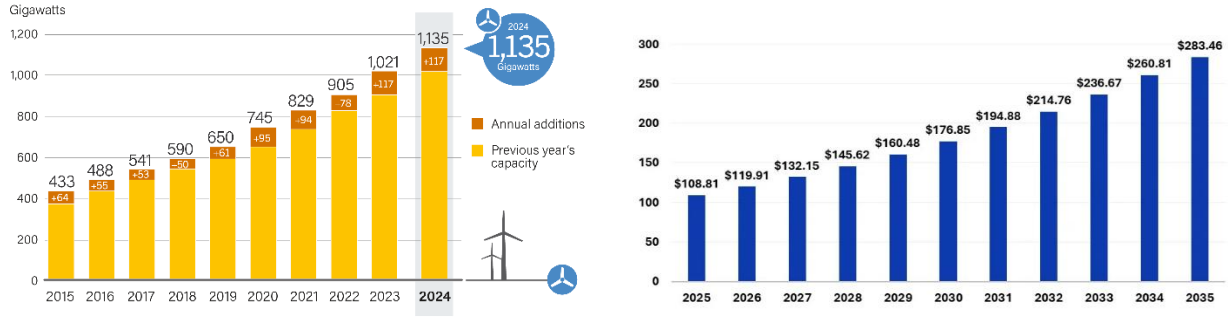


Figure 0.1 Global wind power capacity and Wind energy market size (USD Billion)

of design variables. The Class Shape Transformation (CST) method is useful for this purpose because it can represent smooth aerodynamic shapes using a relatively small number of coefficients. These coefficients can be treated as design variables and modified by an optimization algorithm. Genetic Algorithm (GA), a population-based evolutionary method, is suitable for this

type of problem because it can explore a large design space without requiring gradient information. Its search process is controlled mainly by the selection, crossover and mutation, while the choice of control parameters strongly affects the balance between exploration and convergence [3], [4].

In this research, the NACA 4412 airfoil is used as the baseline profile. Its geometry is parameterized by CST and optimized using GA, while XFOIL is integrated with the optimization process to calculate the aerodynamic coefficients of each generated airfoil. The final optimized profile is then examined using two-dimensional CFD in ANSYS Fluent and compared with the baseline airfoil. The study considers a wind speed of 5 m/s, corresponding to a Reynolds number of approximately 3.42×10^5 for a 1 m chord, and investigates the aerodynamic performance over an angle-of-attack range from -5° to 15°. Finally, the aerodynamic data of the baseline and optimized airfoils are used in QBlade to examine whether the improvement at airfoil level produces a meaningful improvement in wind-turbine power performance.

1.2 Motivation

The coastal and offshore regions of Bangladesh have potential for wind-energy development, but the available wind resource is generally lower than that of many established wind-energy regions. A recent assessment near Saint Martin Island reported an average wind speed of about 4.51 m/s at 50 m height, while a large portion of the observed wind-speed distribution remained around 3.5-5.5 m/s [5]. This creates a practical need for wind-turbine blades that can operate efficiently under low wind-speed conditions.

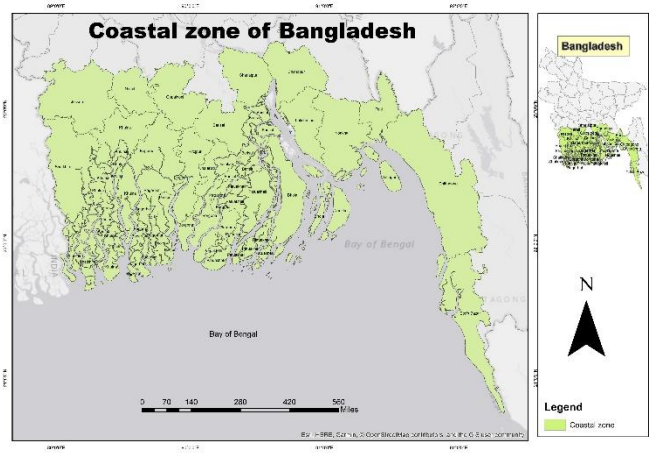


Figure 0.3 Coastal zone of Bangladesh

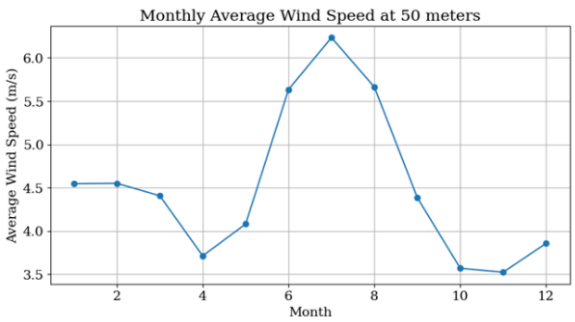


Figure 0.2 Monthly Average Wind Speed Variation in coastal region of Bangladesh

Low wind speed also produces low Reynolds-number flow, where boundary-layer transition and flow separation can strongly influence lift and drag. Previous research has shown that careful modification of airfoil thickness, camber and related geometric features can improve aerodynamic efficiency in the 4-6 m/s range [6]. However, instead of manually modifying only a few geometric quantities, the present work uses CST-based

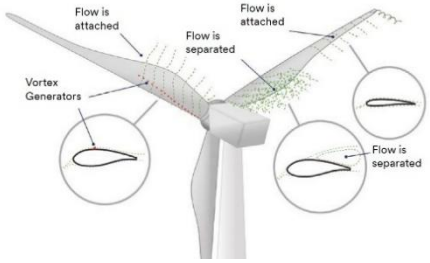


Figure 0.4 Flow separation on turbine blades

parameterization with GA to search a wider range of smooth airfoil shapes. The motivation is therefore to develop an optimized NACA 4412 profile that is more suitable for the low wind-speed conditions relevant to coastal Bangladesh and to verify whether the aerodynamic improvement can also increase turbine-level power performance.



Figure 0.5 Potential wind firm in the coastal region of Bangladesh

1.3 Objectives

The main objectives of this research are:

- To evaluate the aerodynamic performance of the baseline airfoil under the selected low wind-speed operating condition.
- To parameterize the baseline airfoil geometry using the CST method and optimize the CST design variables using a Genetic Algorithm to improve the lift-to-drag ratio.
- To validate and compare the aerodynamic performance of the baseline and optimized airfoils using CFD analysis in ANSYS Fluent.
- To use the aerodynamic data in QBlade and compare the power performance of wind-turbine blades based on the baseline and optimized airfoils.

1.4 Significance of the Research

The significance of this study lies in connecting aerodynamic shape optimization with a regional low wind-speed application. The work combines CST parameterization, GA-based search, XFOIL aerodynamic evaluation, CFD verification and QBlade turbine analysis within one design framework. This allows the optimized geometry to be assessed not only through lift and drag coefficients but also through its influence on turbine power performance. Such a framework can support the development of wind-turbine airfoils for locations where the available wind resource is limited, including the coastal region of Bangladesh. At the same time, the study provides a computational approach that can be extended to other baseline airfoils, operating conditions and optimization objectives.

1.5 Chapter Outline

This thesis is organized into six chapters. Chapter 1 introduces the background, motivation, objectives, and significance of the research. Chapter 2 reviews previous studies on low-wind-speed airfoils, aerodynamic shape parameterization, Genetic Algorithm optimization, and numerical validation, followed by the identification of the research gap. Chapter 3 describes the baseline airfoil and its geometric characteristics; the mathematical formulation and implementation of the CST parameterization method; the Genetic Algorithm optimization scheme and selected GA parameters; the CFD model, mesh, boundary conditions, and numerical validation; and the wind-turbine blade model and QBlade power calculation. Chapter 4 presents and discusses the optimization, aerodynamic, flow-field, and turbine-performance results. Finally, Chapter 5 summarizes the main conclusions, limitations, and possible directions for future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Review

Research on wind-turbine performance has increasingly focused on improving the aerodynamic design of the rotor and its blade sections. A review by Chehouri et al. [7] showed that wind-turbine performance can be improved through several optimization approaches involving the rotor, blade geometry and control system. For blade design, the airfoil is especially important because its lift and drag characteristics directly affect the aerodynamic loading and energy extraction of the rotor. Zhu et al. [8] therefore proposed an integrated airfoil and blade design method for large wind turbines, showing the importance of treating the airfoil and blade geometry as connected design problems rather than independent stages. This idea is also relevant to small turbines, where the airfoil must provide useful lift at lower wind speeds and lower Reynolds numbers.

Low wind-speed operation has been studied from both rotor-level and airfoil-level viewpoints. Wright and Wood [9] investigated the starting and low wind-speed behavior of a small horizontal-axis wind turbine, identifying the starting condition as an important practical issue for small machines. Singh and Ahmed [10] designed and tested a small wind-turbine rotor for low wind-speed applications, confirming that the rotor and its blade sections should be selected according to the local wind regime. These works show that simply applying a blade designed for stronger wind conditions may not provide the best performance when the available wind resource is limited.

The aerodynamic difficulty becomes greater when the Reynolds number is low. Lissaman [11] explained that low-Reynolds-number airfoils are strongly affected by laminar separation, transition and the formation of separation bubbles. Yarusevych et al. [12] experimentally studied

vortex shedding from an airfoil at low Reynolds numbers and showed that the flow structure can change considerably with the operating condition. Because separation reduces lift and increases drag, the geometry of the airfoil becomes very important in this regime. This explains why the maximum thickness, camber and their locations are frequently considered when developing airfoils for low-speed wind turbines.

Several dedicated airfoil families have been developed for wind-energy applications. Tangler and Somers [13] introduced NREL airfoil families for horizontal-axis wind turbines, while Timmer and Van Rooij [14] summarized wind-turbine airfoils developed at Delft University of Technology. For smaller turbines, Giguere and Selig [15], [16] studied low-Reynolds-number airfoils and later developed new airfoils for small horizontal-axis wind turbines. Together, these studies established that airfoils designed for large turbines or high Reynolds numbers cannot always be transferred directly to small or low-speed turbines without considering the change in flow behavior.

As numerical methods became more available, researchers moved from selecting existing airfoils toward modifying or optimizing their shapes. Xudong Wang et al. [17] applied shape optimization to wind-turbine blades, and Henriques et al. [18] used a pressure-load inverse method to design a new airfoil for an urban wind turbine. However, a direct optimization process requires a compact mathematical description of the airfoil. If every surface coordinate is treated as an independent design variable, the search space becomes unnecessarily large and the optimization becomes difficult.

For this reason, airfoil parameterization has become an important part of aerodynamic shape optimization. Samareh [19] reviewed shape-parameterization methods and emphasized that the chosen representation has a strong influence on the efficiency and quality of shape optimization. Different methods have been used, including Hicks-Henne functions, Bézier curves, B-splines and PARSEC [20, 21]. Della et al. [22] coupled PARSEC parameterization with an evolutionary algorithm for airfoil shape optimization, demonstrating how a reduced set of geometric variables can be linked directly with an optimizer. These developments created the basis for more efficient automated airfoil design.

Among the available approaches, the Class Shape Transformation method has become widely used because it can represent smooth aerodynamic shapes using a small number of coefficients. Kulfan introduced the CST formulation as a universal parametric geometry representation method, while Kulfan and Bussoletti [23] presented the underlying class and shape-function concept for aerodynamic components. Wu et al. [24] combined CST with a proper orthogonal decomposition approach for benchmark aerodynamic shape optimization, further showing that CST can be used effectively in modern optimization frameworks. Compared with a direct coordinate-based description, CST reduces the number of design variables while maintaining a smooth airfoil surface, which makes it suitable for population-based optimization.

Genetic Algorithm is one of the population-based methods commonly used for aerodynamic optimization. The basic GA framework was established by Holland and later widely developed for engineering optimization by Goldberg [25], [26]. The algorithm evolves a population of candidate solutions through fitness evaluation, selection, crossover and mutation. Katoch et al. [4] reviewed

GA developments and noted that population diversity and the balance between exploration and exploitation are important for obtaining a reliable solution. The same review also emphasized that control parameters such as population size, selection pressure, crossover and mutation affect convergence and solution quality. For aerodynamic applications, Shahrokhi and Jahangirian [27] combined airfoil parameterization, Navier-Stokes analysis and GA, while Orman and Durmus [28] compared shape-optimization techniques coupled with GA for a wind-turbine airfoil.

A methodology that is particularly close to the present research was developed by Akram and Kim [1]. In their Part I study, CST parameterization was combined with GA and XFOIL for the aerodynamic optimization of the NREL S-821 and RAE-2822 airfoils. The optimization targeted lower drag and higher lift-to-drag ratio, and CFD was used as a further validation step for the NREL S-821 case. Their results reported reductions in drag of about 10% and 12% for the optimized S-821 and RAE-2822 airfoils, with corresponding lift-to-drag improvements of 7.4% and 15.9% at the selected operating point. The same study also showed that CST can reduce the number of geometric design variables and therefore provide a more manageable search space for GA.

In Part II, Akram and Kim applied a similar integrated framework to the NREL S809 wind-turbine airfoil and compared CST with PARSEC parameterization [2]. The CST-based optimized airfoil showed an increase of about 11.8% in lift coefficient and 9.6% in lift-to-drag ratio in the aerodynamic analysis, while the CFD comparison also indicated improvement over the original S809 airfoil. The study used XFOIL inside the optimization loop and RANS-based CFD as a higher-fidelity verification tool. This staged approach is useful because XFOIL provides fast

aerodynamic evaluation for many GA candidates, whereas CFD can be reserved for the final designs where more detailed flow information is required.

Low wind-speed airfoil development has also been investigated without a full CST-GA framework. Sang et al. [6] redesigned the S1010 airfoil for small wind turbines operating at 4-6 m/s using XFLR5 and CFD. They varied geometric features such as maximum thickness and its location, and the final VAST-EPU-S1010 profile achieved its best performance with a maximum thickness of 8% and a thickness position near 20%. Their results showed that changes in airfoil geometry can noticeably improve the lift-to-drag ratio under low-speed conditions. The study is important for the present work because it directly demonstrates the need to adapt an airfoil to the low wind-speed range rather than relying only on standard profiles.

The regional wind condition also supports the need for such optimization in Bangladesh. Islam et al. [5] assessed the offshore wind potential near Saint Martin Island using meteorological and geographic data. They reported an average wind speed of about 4.515 m/s at 50 m height and found that much of the wind-speed distribution remained within a moderate-to-low range. Although their work focused on site assessment rather than blade aerodynamics, the reported wind regime provides a practical operating basis for studying airfoils around 4-5 m/s. More recently, CST and GA have also been used outside wind-turbine applications. Kurkin et al. [29] used CST-based profile representation and GA in an aircraft-propeller optimization framework, which further demonstrates the flexibility of this combination for rotating aerodynamic systems.

2.2 Research Gap

The reviewed literature shows that low-speed wind-turbine design, airfoil parameterization, GA-based optimization and CFD validation have been studied widely. However, their combined application under low-Reynolds-number conditions remains limited.

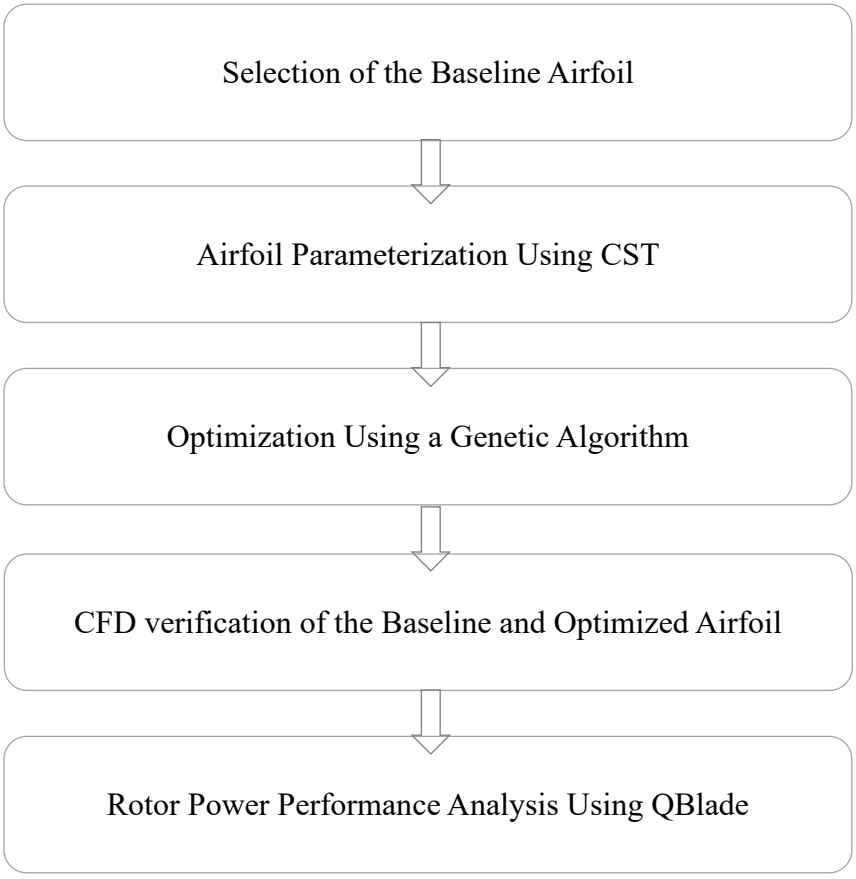
The first research gap is related to the operating condition and optimization approach. The CST-GA studies by Akram and Kim [1], [2] considered NREL S-821, RAE-2822 and S809 airfoils at Reynolds numbers higher than the present value of approximately 0.8×10^{6}. Previously, Sang et al. [6] studied airfoil performance at wind speeds of 4-6 m/s, but they mainly varied selected geometric features rather than using a broader CST-GA optimization approach. Therefore, the application of CST-GA optimization to NACA 4412 under low-speed conditions remains less explored.

The second research gap is the connection between regional wind conditions and airfoil optimization. Islam et al. [5] assessed the offshore wind potential near Saint Martin and identified its wind-speed characteristics, but they did not use these conditions for airfoil or blade design. Thus, there is a need to use the local wind condition as an input for aerodynamic optimization.

The third research gap concerns rotor-level performance assessment. Many studies report improvements only through sectional parameters such as C_L, C_D and C_L/C_D, which do not directly show the effect on complete rotor performance. The present study addresses this gap by comparing the baseline and optimized airfoils using the same QBlade rotor model. In addition, suitable GA control parameters are selected for the current optimization problem because they can strongly influence convergence and solution quality [4].

CHAPTER 3: METHODOLOGY

The methodology follows a systematic process to optimize the NACA 4412 airfoil for low-wind-speed applications. First, the baseline airfoil is selected and its geometry is parameterized using the Class Shape Transformation (CST) method. The CST coefficients are treated as design variables and optimized using a Genetic Algorithm (GA), while XFOIL evaluates the aerodynamic performance of each generated profile. The baseline and optimized airfoils are then compared through CFD analysis in ANSYS Fluent. Finally, their aerodynamic data are applied to the same rotor model in QBlade to examine the effect of airfoil optimization on rotor power performance. The complete sequence of the methodology is summarized as following-



3.1 Selection of the Baseline Airfoil

3.1.1 Airfoil Geometry

1. Chord Line: The chord line is a straight imaginary line that connects the leading edge (front point) to the trailing edge (rear point) of the airfoil. The length of this line is called the chord length, which is a reference dimension for describing the airfoil size.

2. Camber Line: The camber line is the curve that lies exactly halfway between the upper and lower surfaces of the airfoil. It represents the overall curvature of the airfoil and plays a key role in determining how much lift the airfoil can generate.

3. Leading Edge (LE): The leading edge is the front-most point of the airfoil where the airflow first meets the surface. Its shape is very important because it affects how smoothly air flows over the airfoil, especially at different angles of attack.

4. Trailing Edge (TE): The trailing edge is the rear-most point of the airfoil where the upper and lower surfaces meet again. It influences wake formation and drag characteristics of the airfoil.

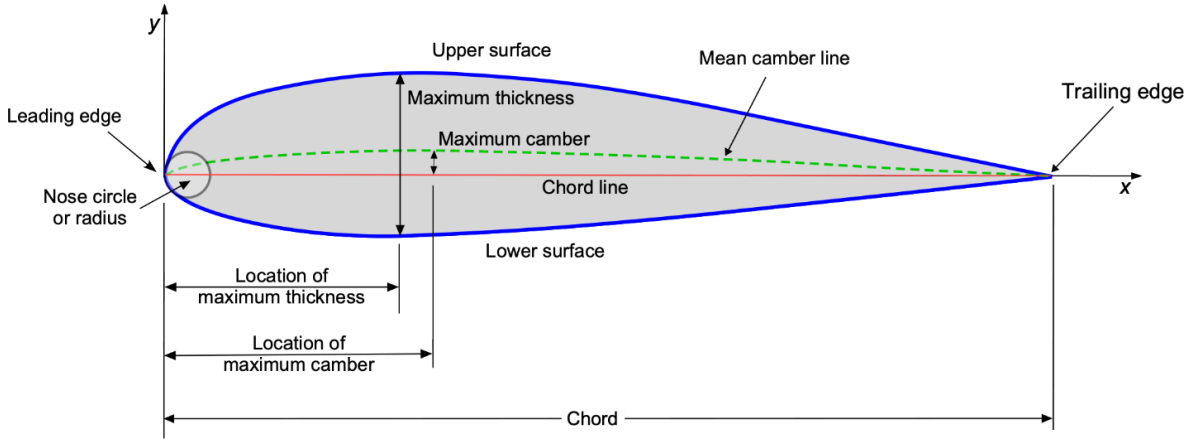


Figure 3. 1 Typical airfoil geometry

5. Thickness: Thickness is the distance between the upper and lower surfaces of the airfoil, measured perpendicular to the camber line. The maximum thickness and its position strongly affect strength and aerodynamic performance.

6. Camber: Camber refers to the maximum distance between the camber line and the chord line. A higher camber generally produces more lift but may also increase drag.

3.1.2 Airfoil Design

The airfoil shape taken in this paper is NACA 4412. It is a standard airfoil designation used to describe its shape in a simple coded form developed by the National Advisory Committee for Aeronautics (NACA). The designation “NACA4412” stands for-

- First digit (4) - Maximum camber of the airfoil is 4% of the chord length. (Camber means the curvature of the airfoil
- Second digit (4) - The position of maximum camber is at 40% of the chord length from the leading edge.
- Last two digits (12**)** - Maximum thickness of the airfoil is 12% of the chord length. NACA 4412 describes a cambered airfoil with moderate curvature and 12% thickness, suitable for good lift generation at low to medium speeds.

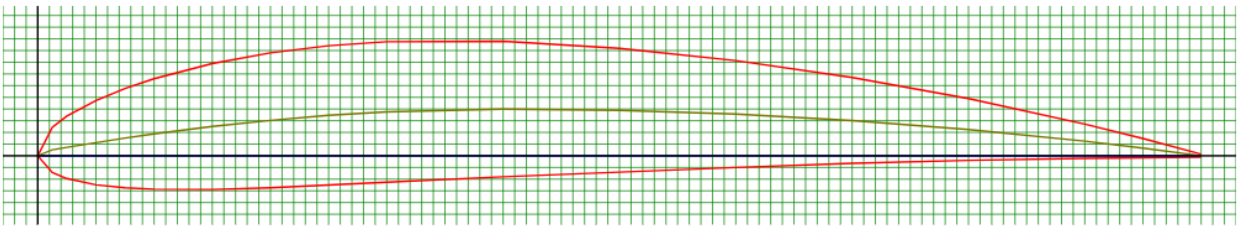


Figure 3. 2 NACA 4412 geometry from UIUC database

3.2 Airfoil Parameterization Using CST

3.2.1 Mathematical Model of CST

The Class Shape Transformation (CST) method is a highly versatile and robust technique for parameterizing both two-dimensional and three-dimensional aerodynamic geometries using Bernstein polynomials. While frequently applied to airfoils, the utility of the CST method extends to a wide array of other aerodynamic structures.

In this approach, the unique geometry of an airfoil is defined by two coefficient arrays, as expressed in Equations (1) and (2). These arrays dictate the curvature coefficients of the upper and lower surfaces, effectively serving as the design variables required for shape optimization. Because of its mathematical flexibility, this parametric scheme can accommodate an extensive variety of airfoil geometries, making it universally applicable.

Mathematically, the CST method relies on two primary components: a class function and a shape function. The class function incorporates fixed parameters that define the general geometric family, while the shape function refines the specific contour. When combining CST with Bernstein polynomials, maintaining the smoothness of the resulting curve is essential for achieving successful and efficient optimization. A detailed mathematical breakdown of the CST framework is presented in the following section.

Equations (1) and (2) shows the relation between the upper and lower surface of CST, where 'C' = class function and 'S' = shape function.

The upper and lower surfaces are expressed as follows:

$$\left(\frac{y}{c}\right)_{upper} = C_{N_2}^{N_1}\left(\frac{x}{c}\right) S_U\left(\frac{x}{c}\right) + \frac{x}{c}\frac{z_{u,TE}}{c} \tag{1}$$

$$\left(\frac{y}{c}\right)_{lower} = C_{N_2}^{N_1}\left(\frac{x}{c}\right) S_L\left(\frac{x}{c}\right) + \frac{x}{c}\frac{z_{l,TE}}{c} \tag{2}$$

where, $z_{u,TE}$ and $z_{l,TE}$ are trailing edge thickness of upper and lower surface, respectively and N 1 = 0.5, N 2 = 1 are constant.

The Class Function defines the general geometric category or the basic family of the shape (for example, a round-nosed, sharp-tailed airfoil shape, or a bluff body). The class function can be expressed by the following equation:

$$C_{N_2}^{N_1}\left(\frac{x}{c}\right) = \left(\frac{x}{c}\right)^{N_1}\left(1 - \frac{x}{c}\right)^{N_2} \tag{3}$$

The Shape Function introduces the specific modifications and local adjustments (using Bernstein polynomials) that differentiate one specific airfoil from another within that category. The general shape function for S_U (for upper surface) and S_L (for lower surface) defines specific shapes within the same class of airfoil as shown in Equations (4) and (5):

$$S_U\left(\frac{x}{c}\right) = \sum_{i=0}^{N_U} W_U(i)\ S\left(\frac{x}{c}, i\right) \tag{4}$$

$$S_L\left(\frac{x}{c}\right) = \sum_{i=0}^{N_L} W_L(i)\ S\left(\frac{x}{c}, i\right) \tag{5}$$

where, W_U, W_L are the upper and lower surface weights respectively. N_U and N_L denote the number of shape functions for the upper and lower surface, respectively. S is a component shape function that is different for the upper and lower surface. In the shape function expression, *S (x/c,i)* is the Bernstein polynomial on order N.

$$S\left(\frac{x}{c}, i\right) = K_i^N * S\left(\frac{x}{c}\right) * \left(1 - \frac{x}{c}\right)^{N-i} \tag{6}$$

where K denotes Binomial coefficient which belongs to order N Bernstein polynomial as follows:

$$K_i^N = \frac{N!}{i!\,(N-i)!} \tag{7}$$

The complete form of equations for both surfaces of airfoils can be expressed as:

$$\left(\frac{y}{c}\right)_{upper} = \left(\frac{x}{c}\right)^{0.5} * \left(1 - \frac{x}{c}\right)^{1.0} \sum_{i=0}^{N_U} \left[W_U(i) * \frac{N_U!}{i!\,(N_U - i)!} * \left(\frac{x}{c}\right)^i \left(1 - \frac{x}{c}\right)^{N_U - i}\right] + \frac{x}{c}\frac{zU_{TE}}{c} \tag{8}$$

$$\left(\frac{y}{c}\right)_{lower} = \left(\frac{x}{c}\right)^{0.5} * \left(1 - \frac{x}{c}\right)^{1.0} \sum_{i=0}^{N_L} \left[W_L(i) * \frac{N_L!}{i!\,(N_L - i)!} * \left(\frac{x}{c}\right)^i \left(1 - \frac{x}{c}\right)^{N_L - i}\right] + \frac{x}{c}\frac{zI_{TE}}{c} \tag{9}$$

The precise geometry of the airfoil surface is governed by the curvature coefficients defined in Equations (8) and (9). To explore new aerodynamic profiles, these baseline curvature coefficients are systematically optimized, yielding a novel set of design variables derived directly from the CST parameterization process.

3.2.2 Implementation of CST Parameterization

To determine the ideal curvature coefficients for the airfoil profile as defined by Equations (8) and (9), a dedicated Python script was developed and executed. The computational workflow for this parameterization process is systematically mapped out in the accompanying flowchart.

The procedure initiates with the retrieval of the raw geometric data, by loading the coordinate file of a baseline NACA 4412 airfoil. Because airfoils possess distinct asymmetric geometries, the script immediately processes these coordinates by segregating them into separate upper and lower surface datasets. Once the surfaces are isolated, the global CST parameters - including the class function exponents (N_1 and N_2) and the order of the Bernstein polynomials are defined.

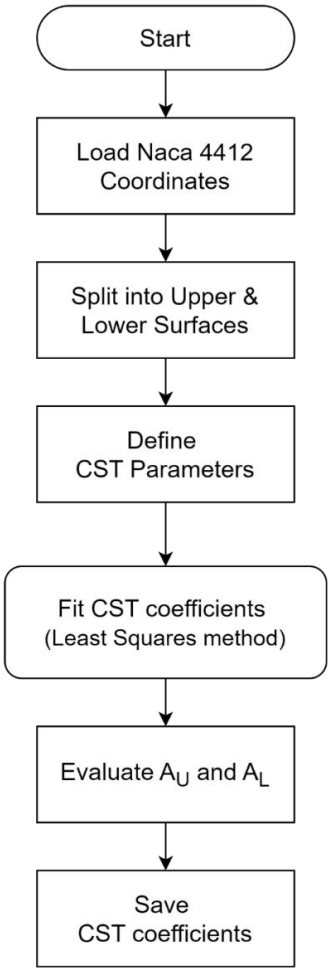


Figure 3. 3 CST parameterization workflow

With the mathematical framework established, a regression analysis is performed. The script fits the CST coefficients to the actual coordinate data points utilizing a Least Squares optimization method, minimizing the residual distance between the source geometry and the parametric curve. This optimization step directly evaluates the final curvature arrays, yielding the specific upper surface coefficients (W_U) and lower surface coefficients (W_L).

Finally, these calculated CST coefficients are saved and exported, serving as the definitive design variables for subsequent aerodynamic optimization stages.

3.2.3 Calculate CST coefficients

The number of coefficients generated will be (N+1). Here, a 5th order polynomial is considered for better results. So, six coefficients for both the upper and lower surfaces. The generated coefficients are presented in Table 3.1

$W_U = [W_{u1}\ W_{u2}\ W_{u3}\ W_{u4}\ W_{u5}\ W_{u6}]$, $W_L = [W_1\ W_{l2}\ W_{l3}\ W_{l4}\ W_{l5}\ W_{l6}]$

Table 3. 1 CST coefficients for optimization

Upper Surface	Lower Surface
WU	WL
0.216	-0.132
0.245	-0.063
0.300	-0.020
0.213	-0.061
0.296	0.005
0.270	-0.020

Figure 3.4 presents an airfoil generated by CST parameterization using 2nd order Bernstein polynomial, where N 1 and N 2 is kept 0.5 and 1.0, respectively [30]. It provides a visual comparison where the initial baseline NACA 4412 geometry is indicated by the solid black line, and the newly parameterized airfoil contour is depicted by the red dashed line.

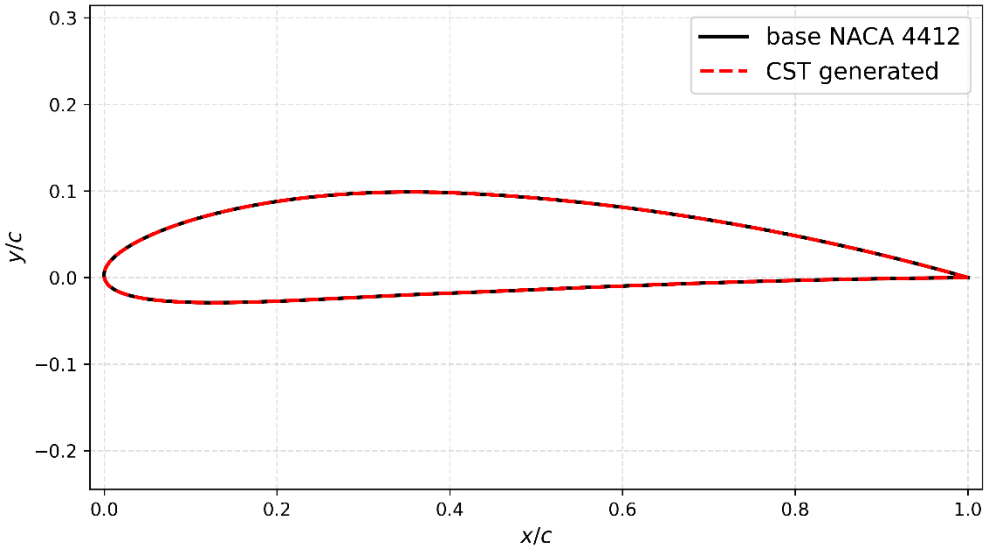


Figure 3. 4 Comparison between baseline and CST generation airfoil

The Figure 3.5 demonstrate how well the CST-generated airfoil matched the baseline airfoil through RMS error for both upper and lower surface, indicating that the selected polynomial degree was sufficiently accurate.

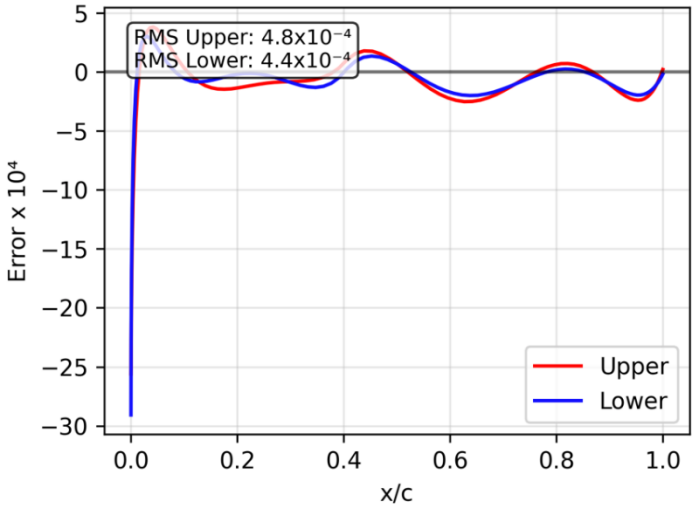


Figure 3. 5 Fitting error distribution

3.3 Optimization Using a Genetic Algorithm

3.3.1 Optimization Scheme

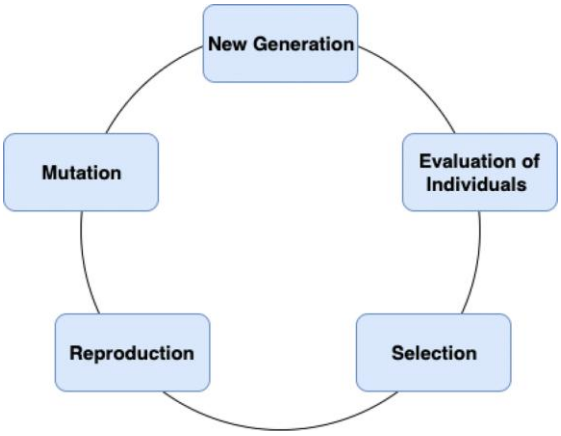
Advanced numerical design techniques play a major role in tackling complicated modern engineering problems. For efficient aerodynamic form optimization, choosing a suitable search mechanism is essential given the wide range of optimization algorithms found in the literature. In this thesis, Genetic Algorithm (GA) is used which is a powerful global optimization technique based on natural selection and biological evolution principles that is ideal for navigating huge, non-linear design domains.

To optimize the profiles of the NACA 4412 profile, the GA was coupled with the XFOIL aerodynamic flow solver via a Python script Pygad module. The optimization routine manipulates the shape curvature coefficients derived as design variables from the Class Shape Transformation (CST) parameterization stage.

While aerodynamic optimization problems can be broadly categorized as either constrained or unconstrained, this work focus on a single-objective formulation aimed at maximizing aerodynamic efficiency. To achieve this, the primary objective is defined by the maximization of the Lift to Drag coefficient ratio (C_L/C_D). The objective function is mathematically expressed as:

$$Maximize: f(x) = Cl/Cd$$

3.3.2 Genetic Algorithm



Unlike traditional gradient-based optimization techniques, the Genetic Algorithm (GA) is a population-based metaheuristic inspired by the mechanisms of natural selection and biological evolution. Initially conceptualized by John Holland in the 1960s, the framework was later expanded and formalized for engineering applications by David E. Goldberg in 1989. As a prominent subset of Evolutionary Algorithms (EAs), GAs are highly effective at navigating complex, non-smooth, and discontinuous design spaces where conventional gradient methods often fail.

The optimization process relies on a suite of bio-inspired operators- specifically selection, crossover, and mutation- to iteratively evolve a population of candidate solutions (chromosomes) [26]. The computational mechanics of the algorithm is shown below:

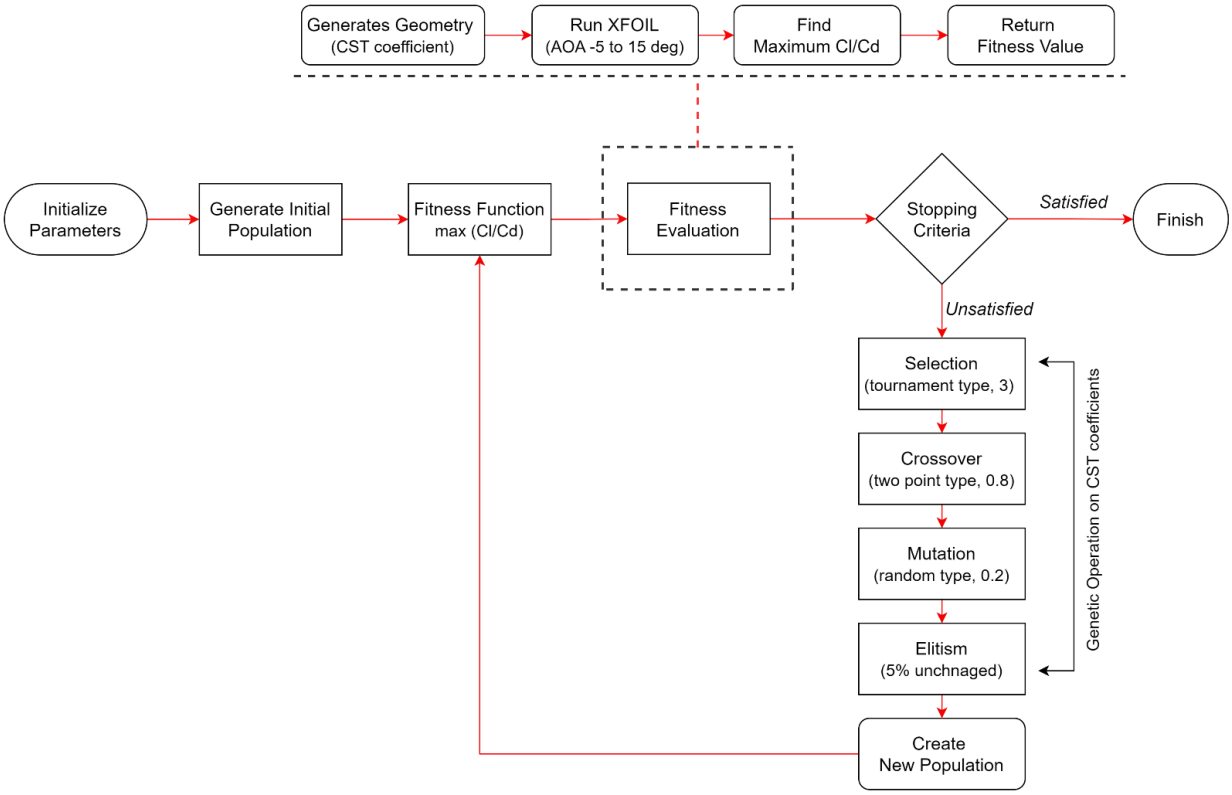


Figure 3. 6 Genetic Algorithm workflow

3.3.3 GA Parameters

Initialization

The initial population generates parameters randomly within defined gene space ranges allowing all the possible solution. Population size (Ps) controls the size of the initial population and it can be determined by the nature of the problem which contains several hundred or thousands of possible solutions.

Number of Parents Mating

This parameter defines how many high-performing chromosomes are selected from the current generation to act as parents. These selected individuals exchange genetic information via crossover to produce the offspring for the next generation.

Selection

Selection is the process of choosing how the better-performing individuals from the current population to act as parents for the next generation. Airfoils with higher fitness values have a greater chance of being selected because they possess more desirable aerodynamic characteristics. This step ensures that useful design features are carried forward during the optimization process.

Crossover

After selection, crossover is performed to create new offspring. During this operation, information from two parent solutions is combined to generate new airfoil designs. The offspring inherit characteristics from both parents, allowing the algorithm to explore new regions of the design space while preserving favorable traits from previous generations.

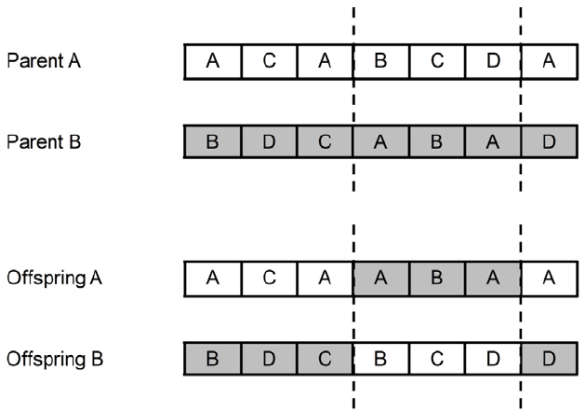


Figure 3. 7 Exchanging genes among parents to form offspring

Mutation

Mutation introduces small random changes to the CST parameters of selected individuals. This step helps maintain diversity within the population and prevents the optimization process from

becoming trapped in a local optimum. By occasionally introducing new design variations, mutation allows the algorithm to continue searching for potentially better solutions. Mutation probability cannot be less than equal to zero and greater than equal to 1. The probability of mutation can be described by relation as shown in Equation (13).

$$Pm = \frac{\sum(bits) + 1}{2 * P_s * \sum(bits)} \tag{10}$$

Elitism

A selection strategy that guarantees the absolute highest-ranking chromosomes (the fittest individuals) from the current generation are passed directly into the next generation completely unaltered. By preventing crossover or mutation from destroying the best solutions found so far, elitism ensures that the algorithm's peak performance never degrades from one generation to the next.

Generation

The newly created offspring replace part or all of the previous population, forming a new generation of candidate airfoils. Their aerodynamic performance is again evaluated, and the cycle of selection, crossover, and mutation is repeated.

The optimization process continues over many generations until a stopping criterion is reached, such as a maximum number of generations or no significant improvement in performance. The airfoil with the highest fitness value at the end of the process is considered the optimized design. By following this evolutionary approach, the Genetic Algorithm efficiently searches for an improved airfoil geometry that provides higher aerodynamic efficiency.

The whole optimization process is summarized in the flow chart below:

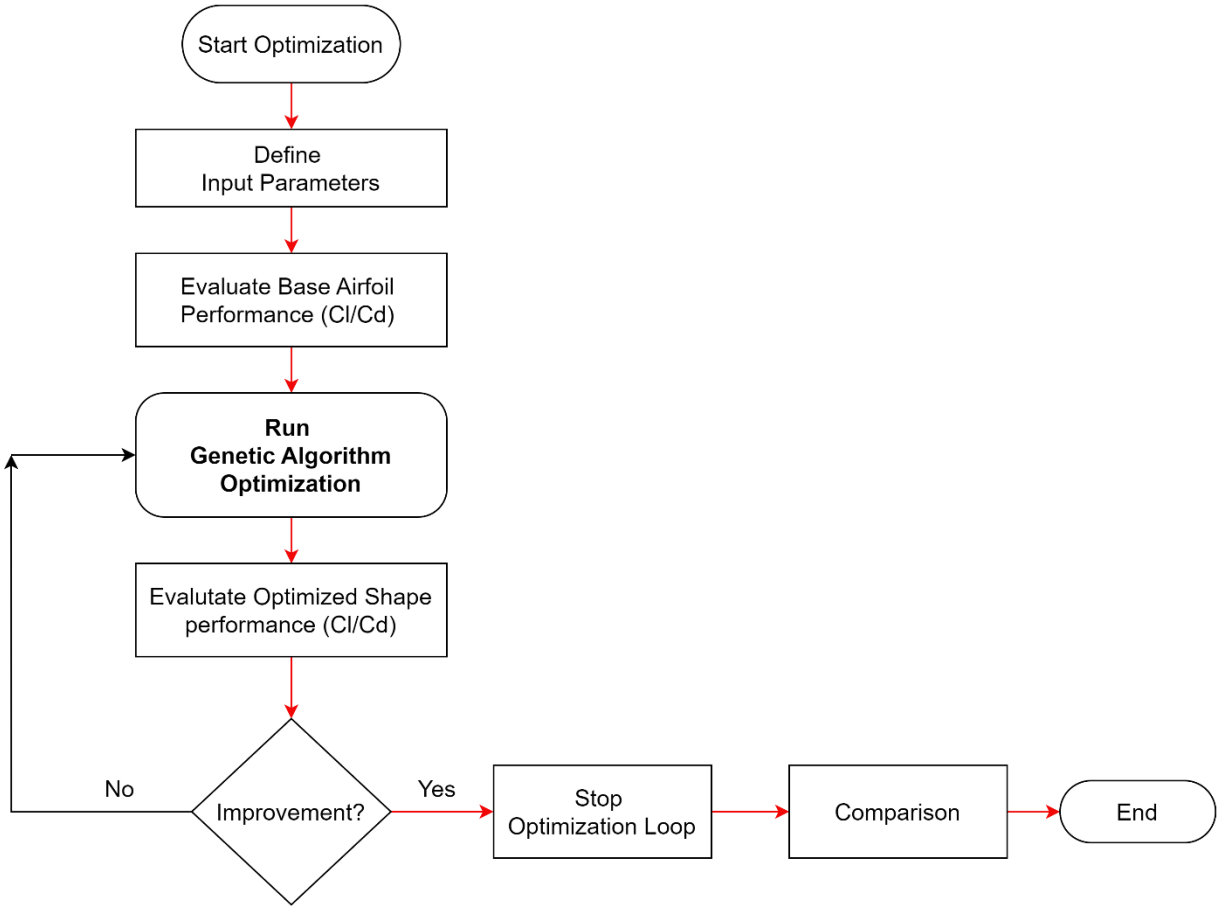


Figure 3. 8 Optimization workflow

3.3.4 Selection of GA parameters

To systematically configure the optimization, the Genetic Algorithm (GA) parameters are categorized into two functional groups: Search space variables and GA operators.

- Search Space Variables**:** This group defines the search boundaries and scale of the optimization environment. It comprises the population size, maximum number of generations, number of mating parents, and explicitly defined gene space (the upper and lower bounds of the design variables).
- GA Operators: This group dictates the evolutionary mechanics used to manipulate the population across iterations. It includes parent selection, crossover, and mutation.

Through an empirical process of testing and tuning various combinations of these parameters, the configuration yielding the highest aerodynamic performance was determined. The selected parameters for this study are summarized in Table 3.2

Table 3. 2 Selected GA parameters

Population size	100	Number of tournaments	4
Generation	150	Mutation type	Random
Number of parents mating	20	Mutation rate	0.05
Gene space	0.09 – 0.11	Crossover type	Uniform
Selection type	Tournament	Crossover probability	0.7

Also 5% elitism is considered from one generation to another.

3.4 Numerical Analysis

3.4.1 Computational model

As additional validation of the resulted airfoil, a two-dimensional CFD simulation of both baseline and optimized airfoil model was conducted in ANSYS Fluent, by solving a compressible form of Reynolds Averaged Navier–Stokes equation (RANS) and continuity equation. The mass and momentum conservation equations are expressed as follows [31]:

$$\frac{\partial}{\partial x_i}(\rho u_i) = 0 \tag{11}$$

$$\frac{\partial(\rho u_i u_j)}{\partial x_j} = \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_i}\left(\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial u_j}{\partial x_i}\right)\right) + \frac{\partial}{\partial x_j}\left(-\rho\overline{u_\iota}\overline{u_J}\right) + \rho f_b \tag{12}$$

3.4.2 Turbulence modeling

The Reynolds-averaging of the Navier-Stokes equations (Eq. 12) introduces the Reynolds stress tensor $\frac{\partial}{\partial x_j}\left(-\rho\overline{u_\iota}\overline{u_J}\right)$, an additional six unknown stress components that render the RANS system mathematically underdetermined. This is the fundamental closure problem: a turbulence model is required to relate these stresses to the computable mean-flow quantities, enabling a closed and solvable system. While Direct Numerical Simulation (DNS) and Large Eddy simulations (LES) resolve turbulent fluctuations with greater fidelity, their computational cost is high. RANS-based modelling thus provides the necessary balance between physical accuracy and computational economy.

At the present operating condition (U_∞ = 5 m/s, Re ≈ 342,370), the boundary layer over the suction surface of the NACA 4412 airfoil does not become turbulent at the leading edge. Instead, a laminar

region persists over a substantial chord fraction before undergoing transition- often manifesting as a laminar separation bubble (LSB)- before reattaching as a turbulent boundary layer. Fully turbulent models such as the standard k–ε and the two-equation k–ω SST impose turbulent production throughout the entire domain from the leading edge, artificially elevating eddy viscosity in the laminar region, suppressing the separation bubble, and producing erroneous lift-to-drag predictions. The standard k–ω SST model is specifically known to over-predict Reynold's shear stresses and delay flow separation on the suction side- a systematic error that directly corrupts the Cl/Cd objective of the present optimization.

The four-equation Transition SST (γ–$Re_{\theta t}$) model of Langtry and Menter (2009) is therefore adopted. By solving additional transport equations for the intermittency factor $\gamma \in [0,1]$ and the transition momentum thickness Reynolds number $\hat{Re}_{\theta t}$, the model explicitly captures the evolution from laminar to turbulent flow and reproduces the laminar separation bubble- a physically critical feature at this Reynolds number that fully turbulent models structurally cannot represent. Multiple validation studies on low-Reynolds-number airfoils have confirmed that the γ–$Re_{\theta t}$ formulation yields the best agreement with experimental transition locations and aerodynamic force coefficients among available RANS closures, and is fully compatible with the local-variable framework of ANSYS Fluent. Accordingly, the Transition SST model is the most physically faithful and numerically appropriate closure for the present low-speed airfoil simulation.

3.4.3 Computational domain

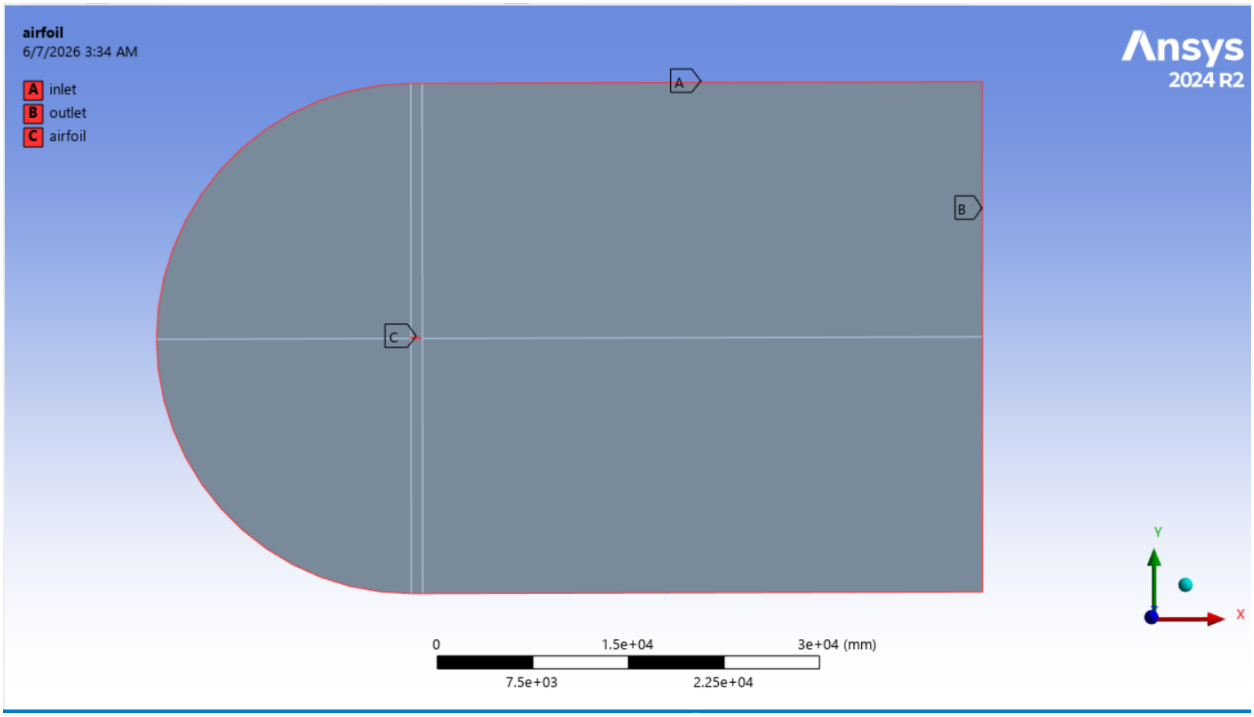


Figure 3. 9 Computational domain blocks

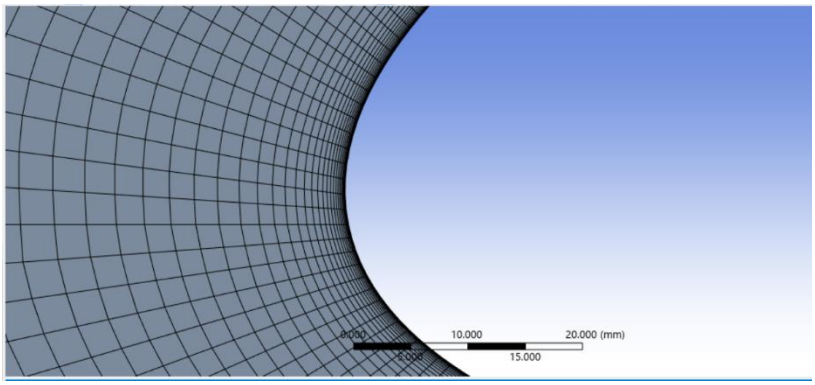


Figure 3. 10 Refined meshing near the airfoil edge

Detailed view of the mesh near the leading edge. The first layer height is set to 0.05mm to keep the y^+ value less than 1. The average aspect ratio of the mesh is 4.1, skewness is 0.089 and orthogonality is 0.96.

3.5 Numerical setup and Boundary conditions

The initial inputs and boundary conditions for the setup of simulation are summarized in Table 3.4. The Reynolds number is considered 0.34 million and was compared with original airfoil data to validate the present simulation. Flow at this Reynolds number and Mach number is treated as incompressible flow.

A Steady state, Pressure based solver type are considered. The Turbulence model is selected as Table 3.3:

Table 3. 3 Turbulence model parameters

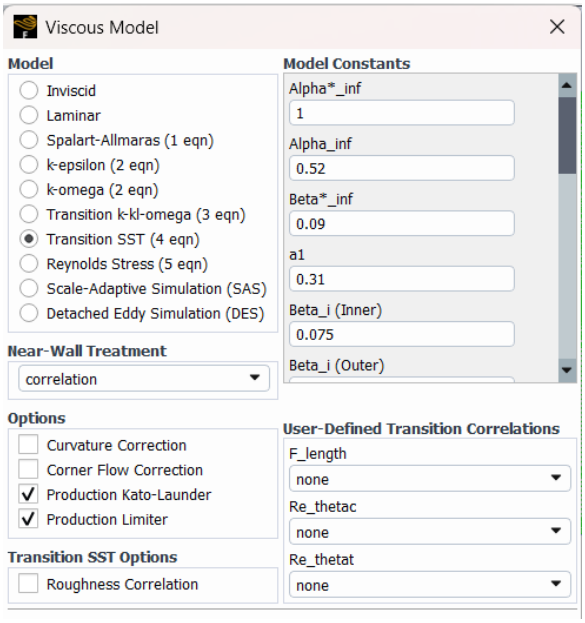


Table 3. 4 Operating parameters for the airfoil

Flow velocity	5 m/s
Working fluid	Air
Chord length	1 m
Reynolds Number	342,370
Dynamic viscosity	1.7894×10^{-5} kg/ms
Air density	1.225 kg/m^3
Temperature (T)	288 K
Pressure (P)	101325 Pa
Angle of Attack (deg)	-5 to 15

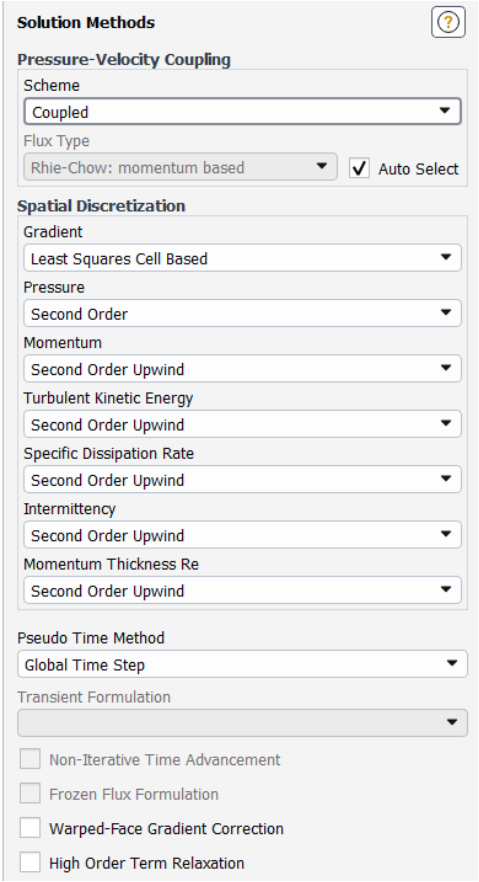
The inlet is velocity inlet and the outlet is pressure outlet. The airfoil is considered as the wall. The Turbulence intensity is 5% and the turbulent viscosity ratio is 10.

Coupled method is chosen for Pressure-velocity coupling. Discretization for Pressure is Second order and Second order upwind for every other setting.

The residuals for continuity, x-velocity, y-velocity, k, omega, intermit and retheta are considered as 1e-06.

Hybrid initialization is chosen and the Time scale factor is 0.05 for stable convergence.

Table 3. 5Solution scheme settings



3.6 Numerical Setup Validation

- To validate the numerical setup, the lift coefficient C_L of the NACA 4412 airfoil was compared against Ansys Fluent CFD results and Baylor University experimental data across a range of angles of attack from −5° to 13°. The Reynolds number is kept constant (Re = 0.15M) for all three cases to maintain the dynamic similarity between these cases.
- The Ansys Fluent result shows very good agreement with the experimental data across the entire range, confirming the reliability of the numerical setup.
- This validated setup is then further used to analyze the optimized airfoil and compare it against the baseline.

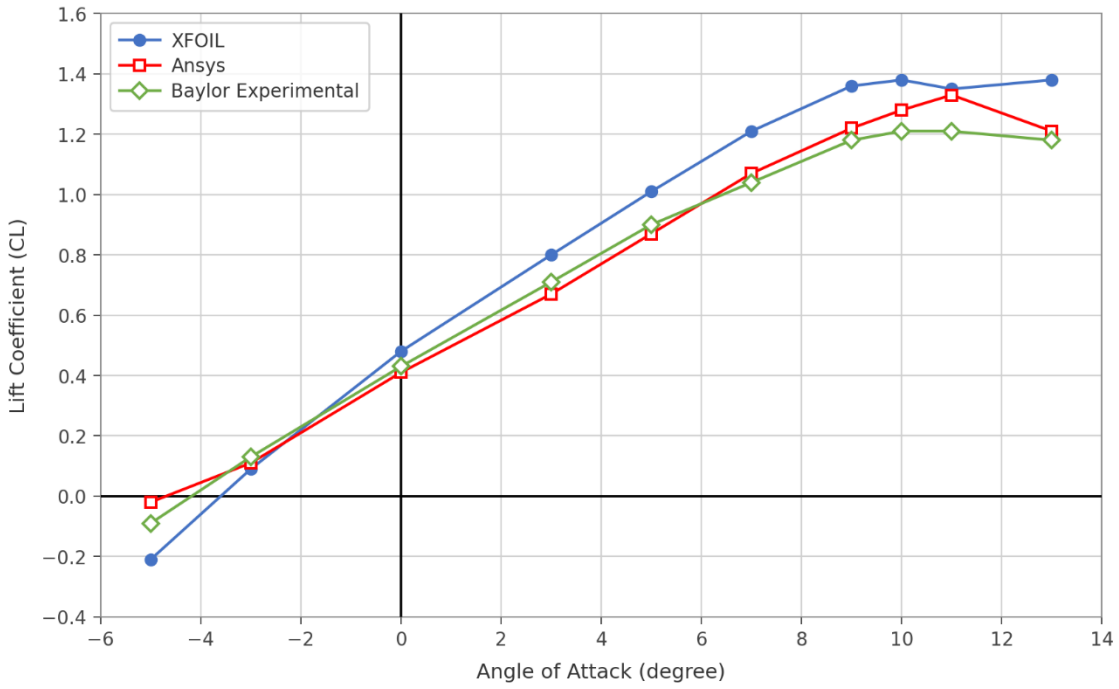


Figure 3. 11 Validation of numerical setup

3.7 Rotor Power Performance Analysis

3.7.1 Background of Power Calculation

The ultimate purpose of improving an airfoil is not only to achieve better aerodynamic characteristics but also to enhance the overall performance of the wind turbine. While parameters such as lift coefficient, drag coefficient, and lift-to-drag ratio provide valuable information about the aerodynamic behavior of an airfoil, they do not directly indicate how much power a wind turbine can generate. Therefore, power calculation becomes an essential step in evaluating the practical benefits of an optimized airfoil design.

The amount of power produced by a wind turbine depends on its ability to extract kinetic energy from the wind. Since the aerodynamic characteristics of the blade airfoil strongly influence this energy extraction process, even small improvements in lift and drag can lead to noticeable increases in power output. An airfoil with a higher lift-to-drag ratio generally enables the turbine blade to operate more efficiently, resulting in improved energy capture, particularly under low wind speed conditions.

In this study, power calculation was performed using QBlade software, which is widely used for the aerodynamic analysis and performance prediction of wind turbines. The aerodynamic data obtained from the baseline NACA 4412 and optimized airfoil were incorporated into the software to evaluate their influence on turbine performance. By comparing the power output generated by both airfoils under the same operating conditions, it was possible to determine whether the optimization process resulted in a meaningful improvement in energy production.

Since the coastal regions of Bangladesh experience relatively low wind speeds, maximizing power extraction is particularly important. Therefore, the power calculation serves as the final measure of the effectiveness of the airfoil optimization and provides a direct assessment of its potential contribution to wind energy generation in low wind speed environments.

3.7.2 Blade Geometry

The geometry of a wind turbine blade plays a crucial role in determining how efficiently the turbine can capture energy from the wind. A typical wind turbine blade is not uniform along its length; instead, its shape gradually changes from the root to the tip to achieve the best aerodynamic performance and structural strength.

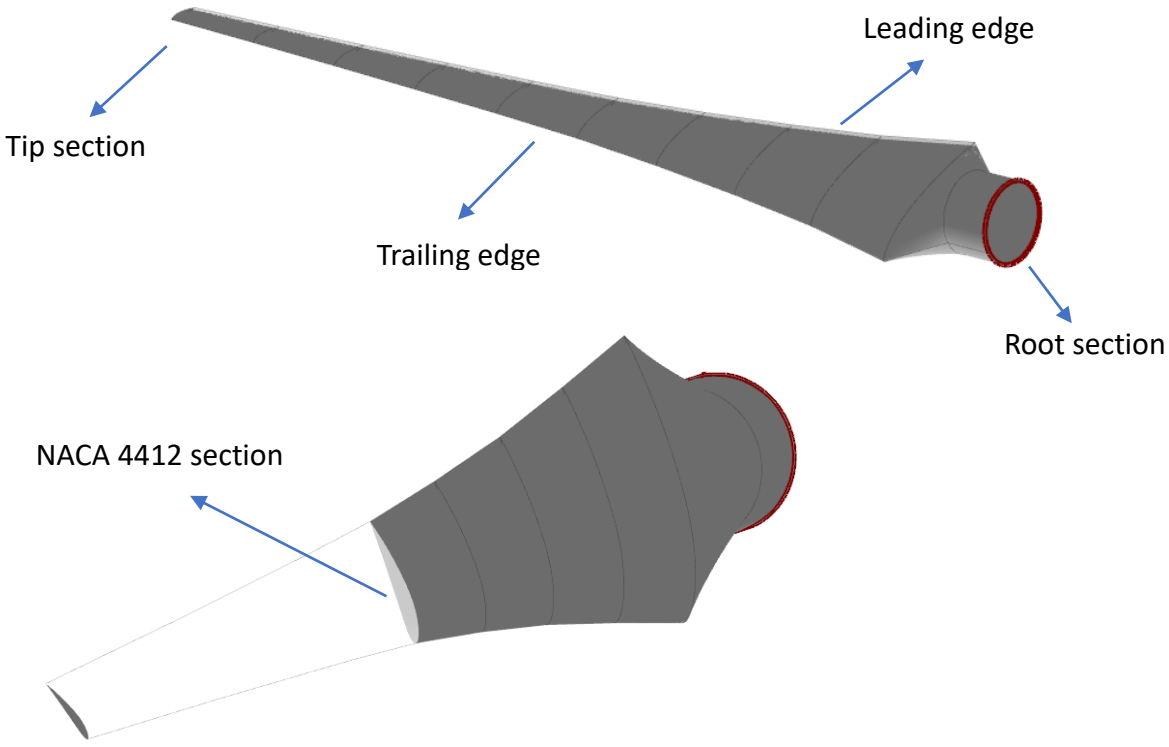


Figure 3. 12 Turbine blade geometry

Root Section

The root section is the part of the blade that is attached directly to the hub of the wind turbine. This region experiences the highest structural loads and bending forces. Therefore, the root section is usually thicker than the rest of the blade to provide sufficient strength and stiffness. Since the rotational speed near the hub is relatively low, the root section contributes less to power generation compared to other parts of the blade.

Tip Section

The tip section is the outermost part of the blade and moves at the highest rotational speed. This region contributes significantly to the extraction of wind energy and has a major influence on the overall efficiency of the turbine. To reduce drag and improve aerodynamic performance, the tip section is generally thinner and more streamlined than the root section.

Leading Edge (LE)

The leading edge is the front edge of the blade that first comes into contact with the incoming airflow. It is designed with a smooth and rounded shape to guide the air smoothly over the blade surface. The condition and shape of the leading edge greatly affect lift generation and flow separation characteristics.

Trailing Edge (TE)

The trailing edge is the rear edge of the blade where the airflow from the upper and lower surfaces meets and leaves the blade. It is usually much thinner than the leading edge. The shape of the trailing edge influences wake formation, drag, and overall aerodynamic efficiency.

Chord Distribution

The chord length is the distance between the leading edge and trailing edge of a blade section. In most wind turbine blades; the chord length is larger near the root and gradually decreases toward the tip. This variation helps maintain efficient aerodynamic loading along the blade span.

Twist Distribution

Wind turbine blades are also twisted from root to tip. The blade angle is larger near the root and gradually decreases toward the tip. This twist ensures that each blade section operates at a near-optimum angle of attack despite the difference in rotational speed along the blade length.

3.7.3 Construction of blades

To analysis in QBlade software, at first sample turbine blades are constructed using the baseline NACA 4412 and optimized airfoil shape. The constructed blade profile is shown in Figure 3.13. A standard chord and twist distribution is taken for designing of the blades and considered same for both cases, presented in Table 3.6

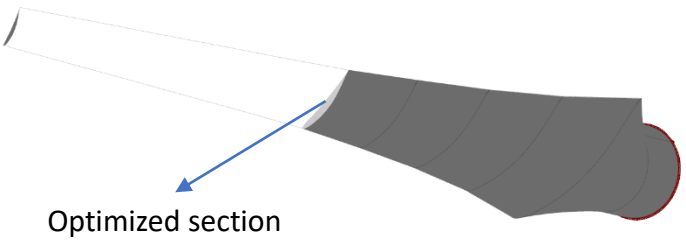


Figure 3. 13 Blade geometry for optimized section

Table 3. 11 Normalized chord distribution

Normalized Chord Distribution	
r/R	c/R
0.1	0.177
0.2	0.121
0.3	0.097
0.4	0.079
0.5	0.065
0.6	0.057
0.7	0.051
0.8	0.045
0.9	0.040
1.0	0.035

After then a sample turbine model with 3 blades for both case is constructed, Figure 3.14. The design parameters, given in Table 3.7, are kept same for both to do a fair comparison. The QBlade software works based on Blade Element Momentum Theory (BEMT). The Schmitz optimization method was used to optimize the blade sections.

Table 3. 7 Design parameters for power calculation

Parameters	Value
Number of Blade	3
Hub Radius (m)	0.06
Rotor Radius (m)	1.06
Blade Length (m)	1
Tip Speed Ratio (TSR)	4
Wind Speed (m/s)	5
Density of Air (kg/m^3)	1.225

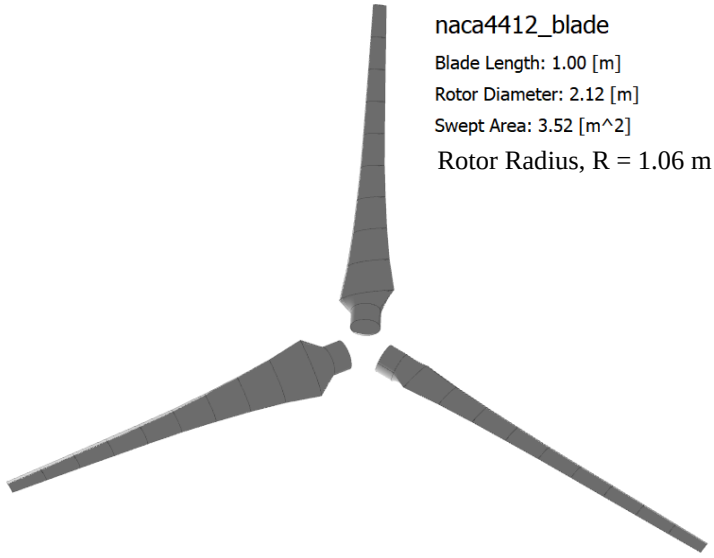


Figure 3. 14 Three blade turbine model

CHAPTER 4: RESULT AND DISCUSSION

4.1 GA convergence and selected configuration

The optimization used the twelve CST coefficients from Chapter 3 as shape variables and maximized the lift-to-drag coefficient ratio (C_l/C_D) evaluated with XFOIL. After testing and tuning various combinations of the GA parameters, the configuration yielding the highest fitness function is determined. The selected parameters are-

Population size = 100

Generation = 150

Parent selection method = Tournament

Mutation type = Random

Crossover type = Uniform

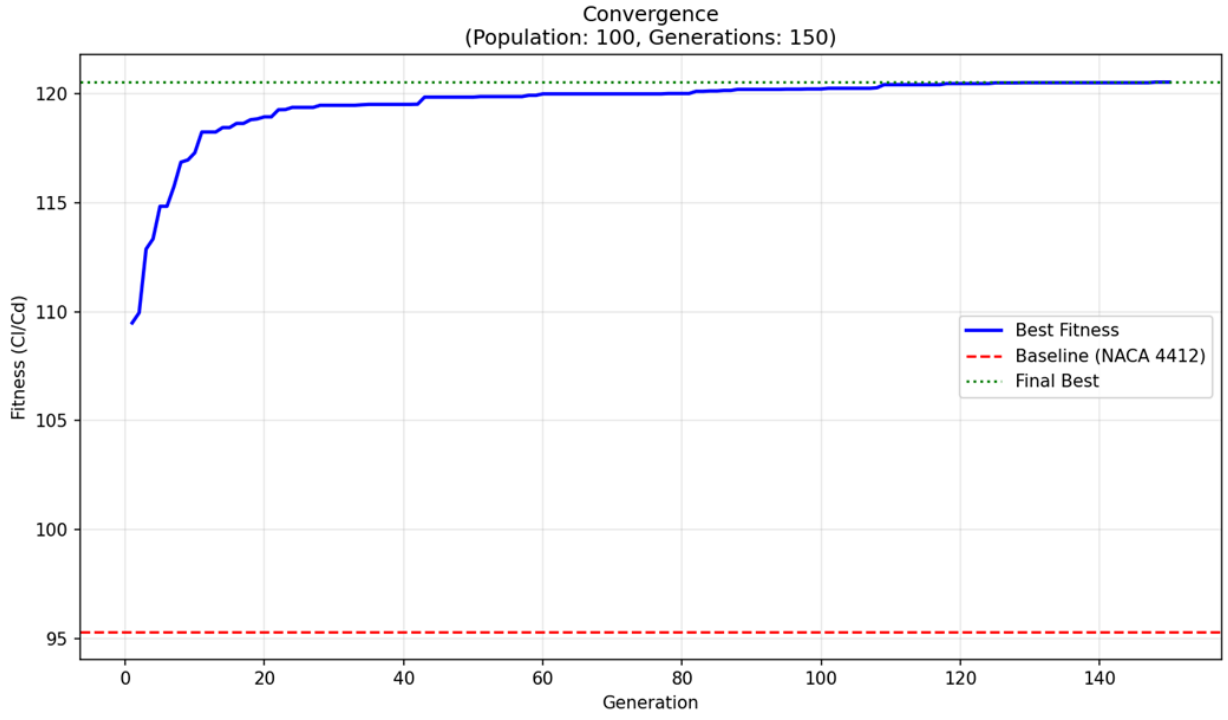


Figure 4.1 Convergence plot

Figure 4.1 shows a rapid increase in the best fitness during the early generations, followed by smaller gains. The curve approaches a nearly constant value toward the end of the 150-generation run. The final fitness is about 119.93, compared with 95.71 for the baseline.

4.2 Geometry comparison

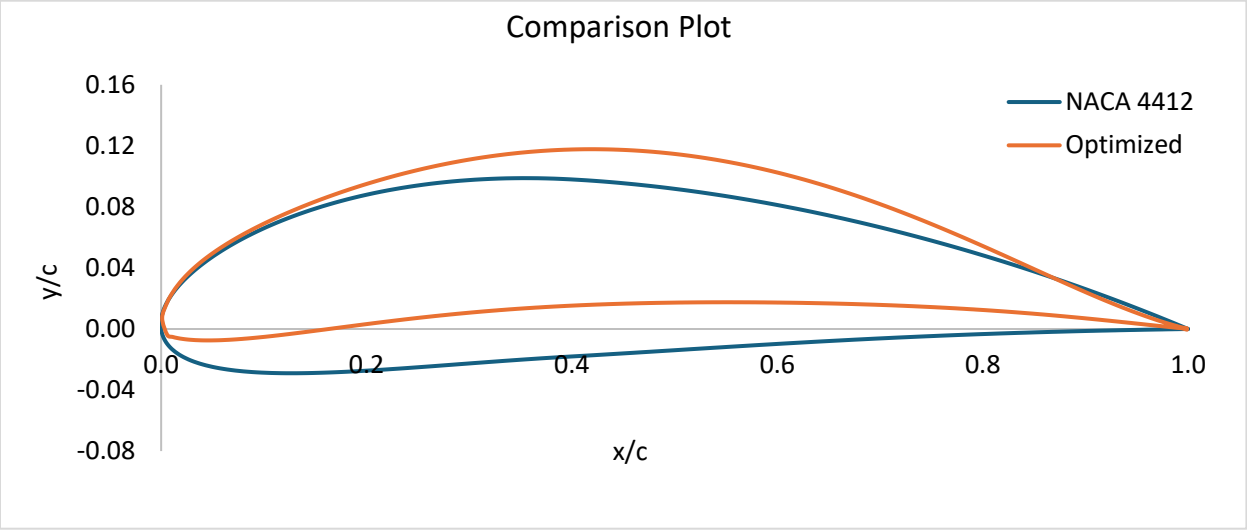


Figure 4.2 Geometry comparison of baseline and optimized airfoil

The optimized profile in Figure 4.2 retains a smooth airfoil outline but has a fuller camber and a smaller thickness than NACA 4412. Table 4.1 shows that the maximum camber rises from 4.0% to 6.7% of chord, and its position moves from 40.0% to 44.1% chord. Maximum thickness decreases from 12.0% to 10.25% of chord and moves from 30% to 38% chord. Thus, the GA selected a more cambered and thinner section while searching for a higher C_l/C_D at the target condition.

Table 4. 1 Comparison of geometric parameters

Airfoil	Thickness (%)	at (%)	Camber (%)	at (%)
NACA 4412	12	30	4	40
Optimized	10.25	38	6.7	44.1

4.3 Aerodynamic performance comparison

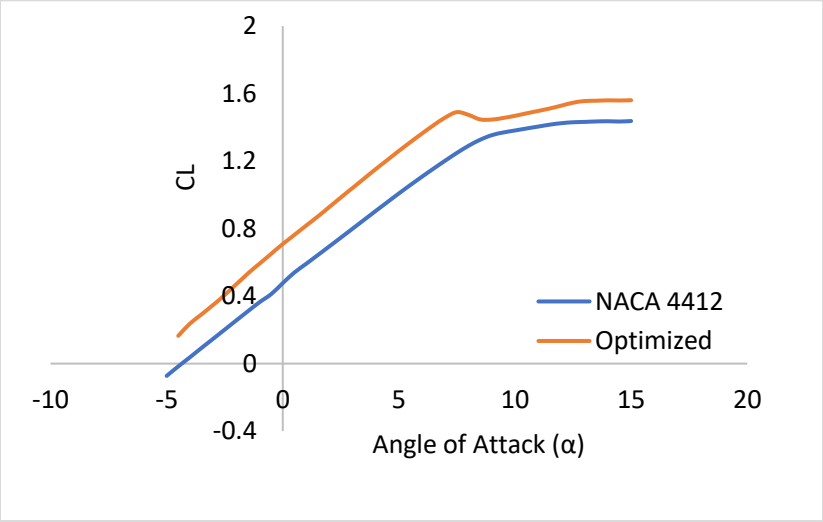


Figure 4.3 Lift coefficient comparison of baseline and optimized airfoil

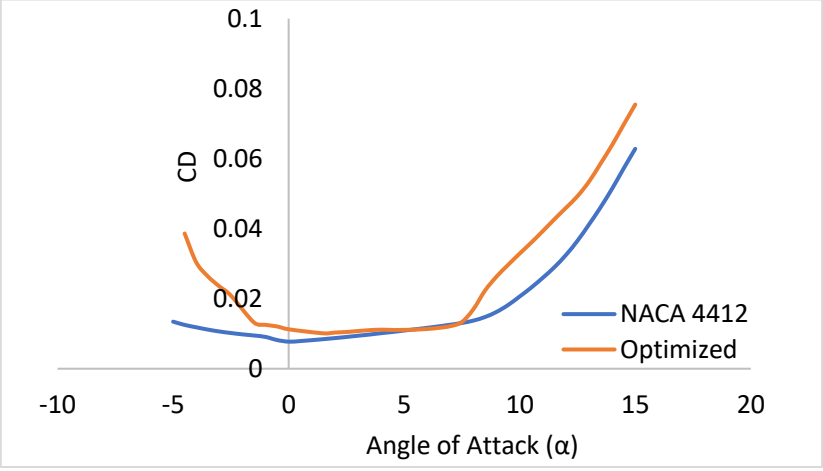


Figure 4.4 Drag coefficient comparison of baseline and optimized airfoil

Across the plotted range from −5° to 15°, the optimized airfoil produces a higher lift coefficient than the baseline (Figure 4.3). Its higher camber also shifts the lift curve upward near zero angle of attack. At higher angles, both curves flatten.

The drag curves in Figure 4.4 shows the main trade-off. Around the low-drag region, the differences are modest. At negative and higher positive angles, the optimized section generally has more drag, with the separation becoming clear beyond about 8°. Higher lift therefore does not imply lower drag over the full operating range.

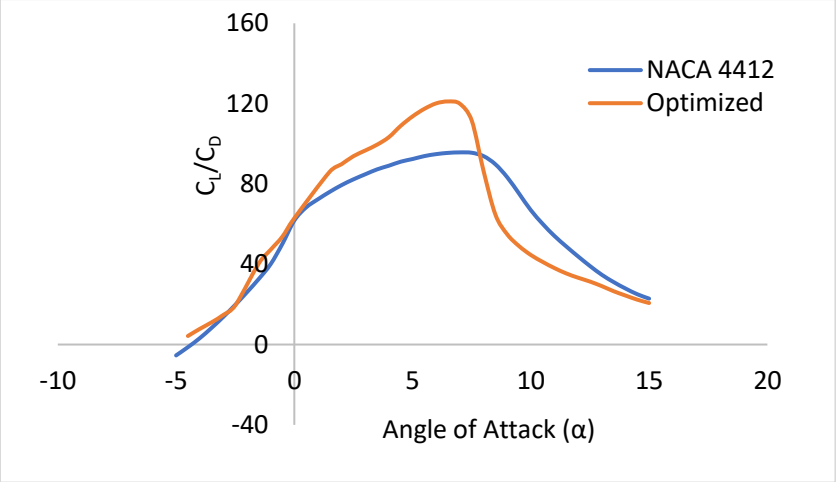
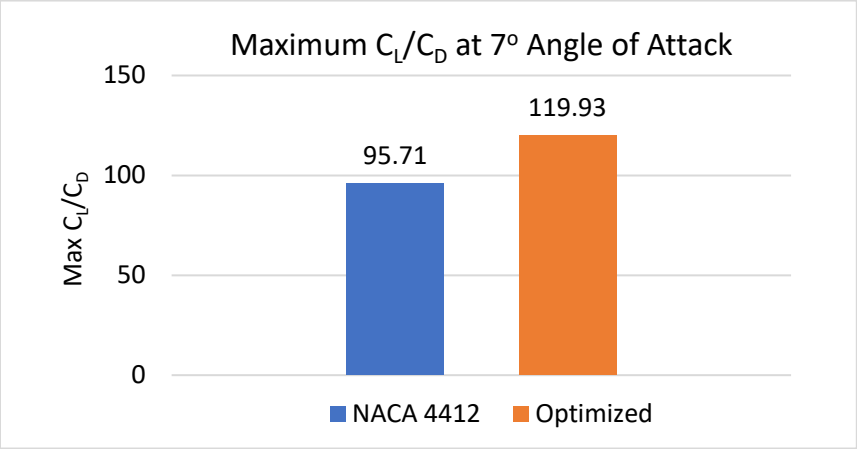


Figure 4.5 Lift to Drag coefficient ratio comparison of baseline and optimized airfoil

Figure 4.5 shows a clear gain in C_l/C_D near the selected design angle of 7°. At this angle the optimized section reaches a max fitness of 119.93, compare to the baseline of 95.71. The relative increase is approximately 25.3%. Above roughly 8°, the optimized ratio falls sharply and the baseline becomes more efficient. This narrower high efficiency region matters if blade sections operate away from their intended angle because of changes in wind speed, rotation speed, or blade position.



The bar chart comparison shows that the Optimized foil has maximum fitness value of 119.93 and around 25.3% increase over the baseline NACA 4412 airfoil.

4.4 CFD verification and Flow pattern comparison

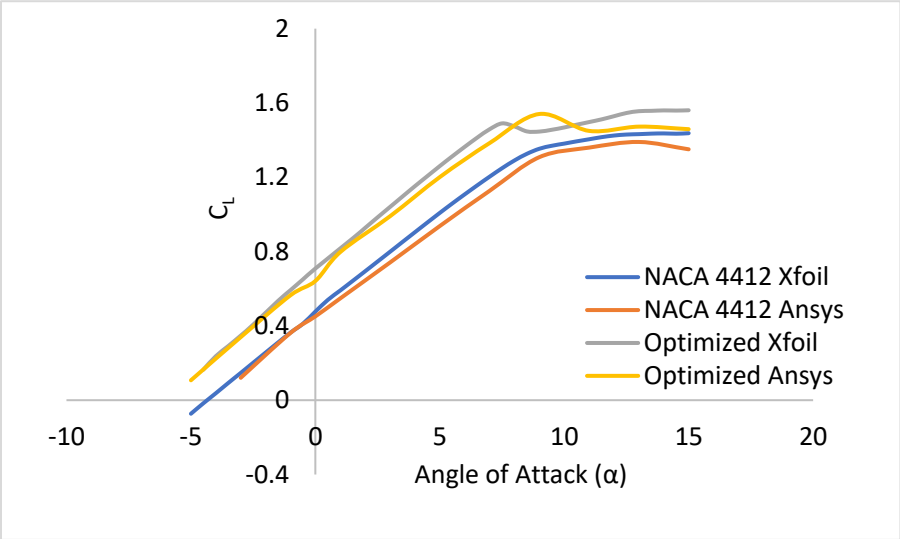


Figure 4.6 Lift coefficient predicted by XFOIL and ANSYS Fluent for both sections

The analysis of both XFOIL and ANSYS Fluent are compared in Figure 4.6, which supports the same ordering in lift: the optimized airfoil has the larger C_l over the displayed angles in both methods. The curves also differ, especially near the upper end of the angle range. For the baseline, the two predictions are comparatively close; for the optimized airfoil, Fluent predicts higher lift over part of the positive-angle range. The results show very close agreement between XFOIL and

Ansys Fluent for both airfoils, further reinforcing the validity of using XFOIL as the aerodynamic evaluator within the genetic algorithm optimization framework.

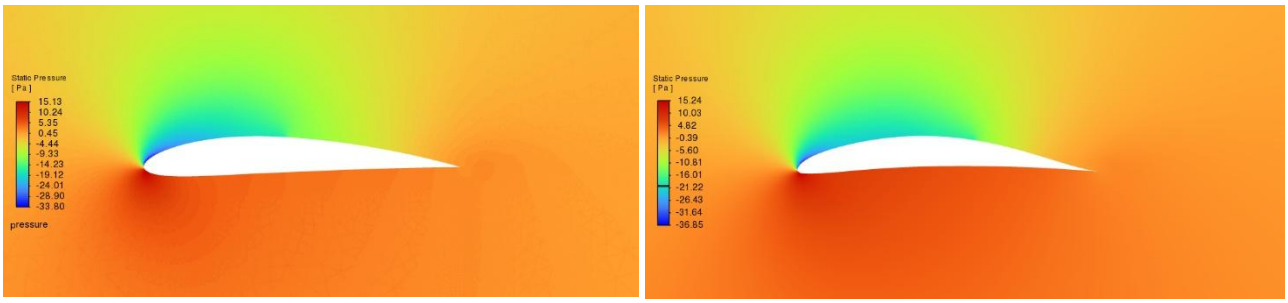


Figure 4.8 Static-pressure fields of NACA 4412 (left) and the optimized section (right) at 7º AOA

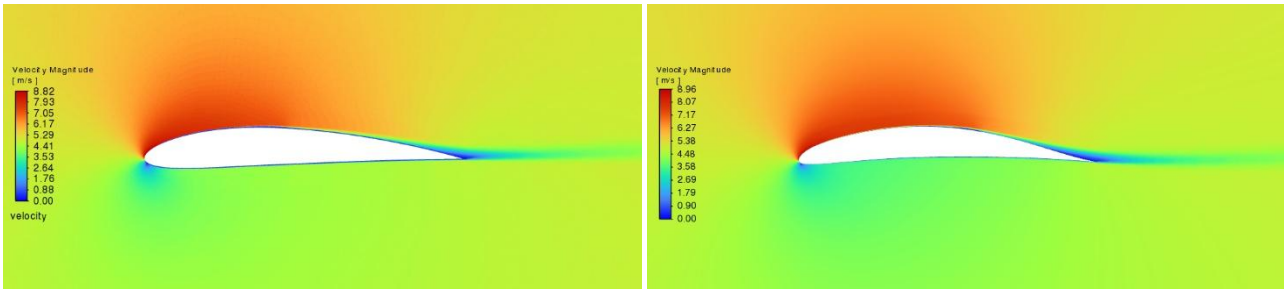


Figure 4.7 Velocity-magnitude fields of NACA 4412 (left) and the optimized section (right) at 7º AOA

In Figures 4.7 and 4.8, both airfoils show lower pressure and faster flow above the upper surface than below it. This pressure difference is consistent with positive lift. The optimized profile changes the size and location of the upper surface accelerated region.

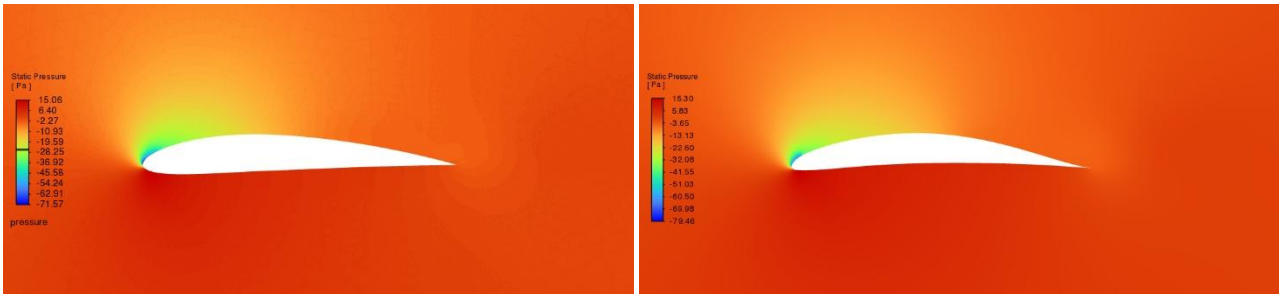


Figure 4.10 Static-pressure fields of NACA 4412 (left) and the optimized section (right) at 13º AOA

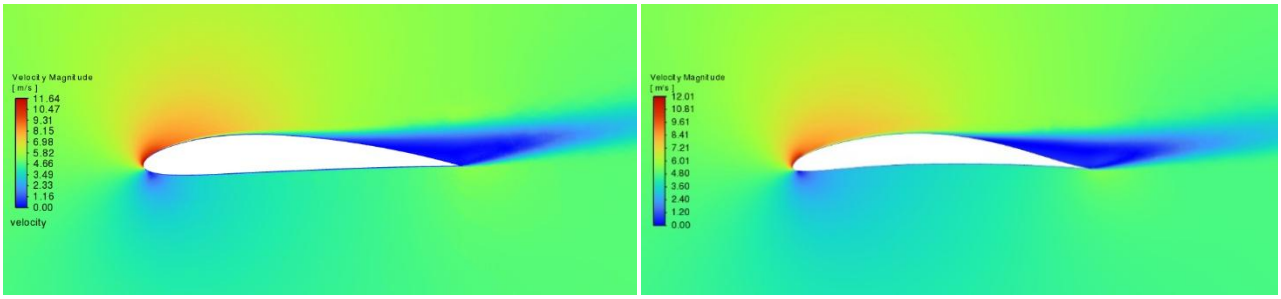


Figure 4.9 Velocity-magnitude fields of NACA 4412 (left) and the optimized section (right) at 13º AOA

At 13° (Figures 4.9 and 4.10), the upper surface low-pressure region and the low-speed wake differ from those at the lower angle. The velocity fields show a more pronounced wake downstream of each airfoil. These images are consistent with changing flow behavior at high angle of attack.

4.5 QBlade Power analysis

Section 4.5 describes the QBlade comparison for two three-bladed rotors with the same geometry: rotor radius 1.06 m, hub radius 0.06 m, and the same chord and twist distributions. The airfoil data are the intended difference between the two cases. Figure 4.11 presents the predicted power coefficient against tip speed ratio (TSR) at a wind speed of 5 m/s.

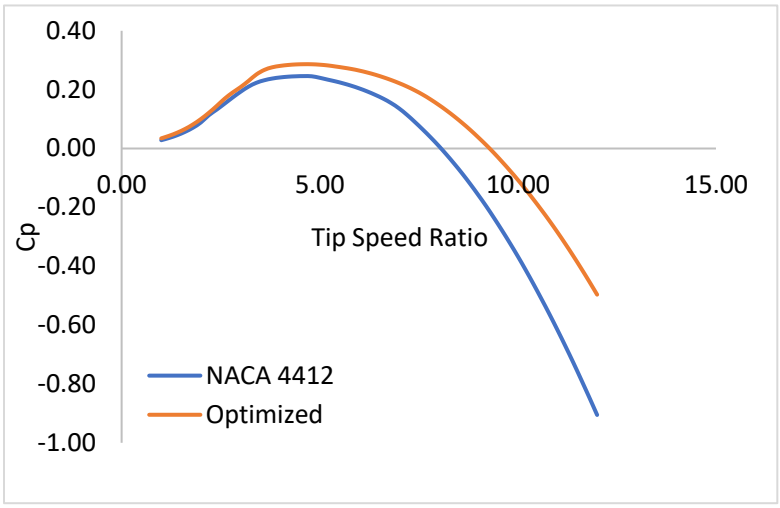


Figure 4.11 QBlade power coefficient versus tip speed ratio for baseline and optimized blade section

Near the selected TSR of 4, the optimized rotor has a higher power coefficient, around 16.67% increase in C_p at this point. Both curves peak around TSR 4–5, and the optimized curve remains higher across much of the plotted range. At very high TSR, both curves become negative. They do not establish useful power generation at those conditions.

Figures 4.12 and 4.13 show higher predicted thrust and torque coefficients for the optimized rotor around TSR 4. The higher torque is consistent with the higher C_p at the same TSR. Higher thrust also means a larger aerodynamic load on the rotor, so it should be considered together with the thinner optimized airfoil in a later structural assessment.

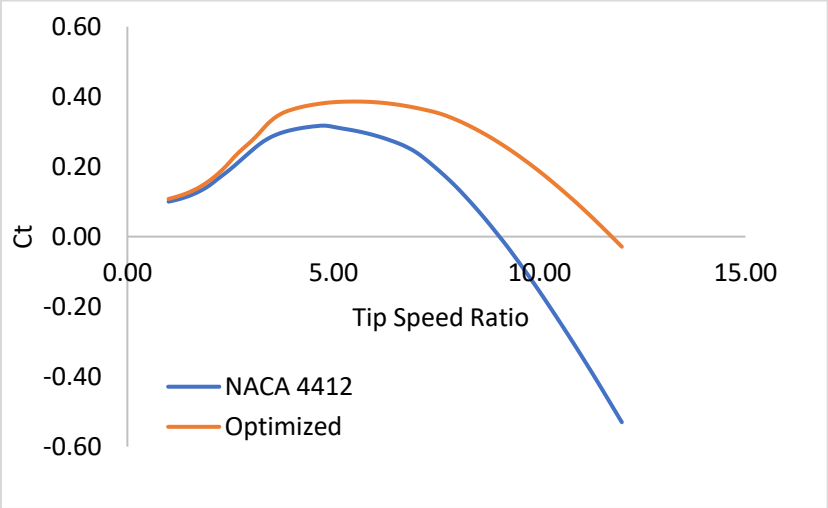


Figure 4.12 QBlade thrust coefficient versus tip speed ratio

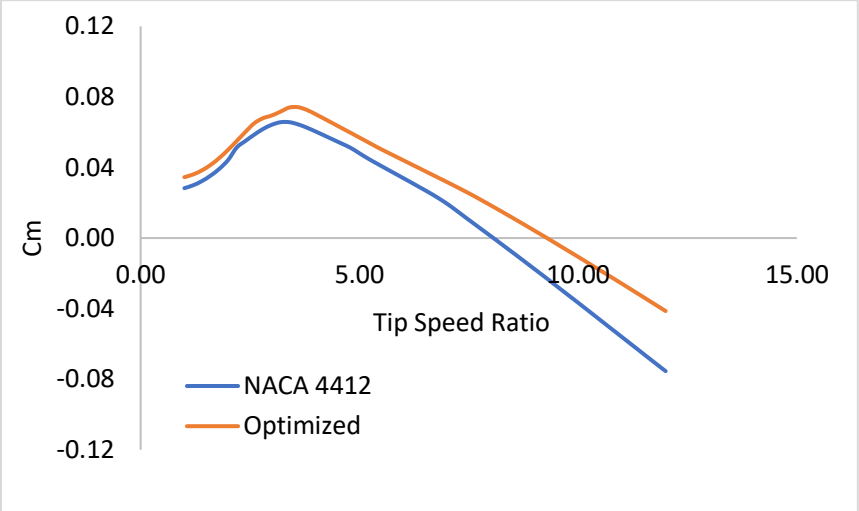


Figure 4.13 QBlade torque coefficient versus tip speed ratio

Overall, the figures support a targeted aerodynamic improvement: the optimized airfoil has higher C_l/C_D at 7° and the rotor model predicts higher C_p at TSR 4. The gain is accompanied by greater camber, reduced thickness, higher drag at larger angles, and greater predicted rotor thrust. The result is therefore most clearly supported for the stated design condition.

CHAPTER 5: CONCLUSION AND FUTURE SCOPES

5.1 Key Findings

This study developed an aerodynamic optimization framework for the NACA 4412 airfoil under the low-wind-speed conditions relevant to the coastal region of Bangladesh. The baseline geometry was represented by twelve Class Shape Transformation coefficients and optimized using a Genetic Algorithm. XFOIL was used within the optimization loop to evaluate each generated profile at a wind speed of 5 m/s and a Reynolds number of approximately 3.42×10^5. The baseline and optimized airfoils were then compared through CFD analysis using ANSYS Fluent. Finally, their aerodynamic data were applied to the same three-bladed rotor model in QBlade to determine whether the sectional improvement produced a gain in rotor performance.

The main findings obtained from the aerodynamic and rotor analyses are summarized below:

- The selected GA configuration converged to a maximum lift-to-drag ratio of 119.93 at an angle of attack of 7°, compared with 95.71 for the baseline NACA 4412. This represents an improvement of approximately 25.3% at the design point.
- The optimized airfoil became more cambered and thinner. Its maximum camber increased from 4.0% to 6.7% of the chord, while the maximum thickness decreased from 12.0% to 10.25% of the chord.
- The optimized section produced a higher lift coefficient across the investigated angle-of-attack range. However, its drag increased at larger positive angles, and the baseline airfoil became more efficient above 8° angle of attack. The improvement is therefore strongest near the selected design condition.

- The ANSYS Fluent results supported the lift trend predicted by XFOIL. The pressure and velocity contours also showed changes in the upper-surface acceleration and wake pattern caused by the modified geometry.
- In QBlade, the rotor using the optimized airfoil achieved an approximately 16.67% higher power coefficient at a tip speed ratio of 4. It also produced higher torque and thrust coefficients around this operating point.

5.2 Limitations and Future works

The study demonstrates the potential of airfoil shape optimization to improve the aerodynamic performance of small-scale offshore wind turbines under low-wind-speed conditions. The optimized airfoil showed improved aerodynamic performance compared with the baseline airfoil, indicating that systematic shape optimization can be useful for developing more efficient airfoils for coastal wind energy applications.

However, the optimization mainly focused on maximizing the lift-to-drag ratio at a single design condition. Therefore, the optimized airfoil may not provide the same improvement at other design or operating conditions. The study was also limited to XFOIL and two-dimensional steady CFD analysis, while three-dimensional effects such as spanwise flow, tip vortices, induced drag, and rotational effects were not captured, which may influence flow separation, aerodynamic coefficients, and overall rotor performance. Moreover, its reduced thickness and higher rotor thrust may create structural and fatigue concerns. Future studies should consider multi-point and multi-objective optimization with structural constraints, followed three-dimensional transient CFD and structural analysis. Further work may also optimize the blade chord and twist distributions and estimate annual energy production using the wind conditions of the selected coastal region.

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